

FROM CASH TO CASHLESS: EVALUATING PAYNOW AND CASH DEMAND IN SINGAPORE[†]

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ABSTRACT

This paper examines whether PayNow reduced aggregate cash demand in Singapore. Using a constructed monthly currency in circulation to gross domestic product (CIC/GDP) ratio series, this paper estimates seasonal autoregressive integrated moving average (SARIMA) interrupted time-series models over the period from April 2014 to December 2019, treating August 2017 as the PayNow intervention month. The candidate models incorporate both step and ramp specifications to compare and distinguish an immediate level shift from a gradual post-intervention decline effect. The selected baseline model estimates a PayNow step coefficient of -0.00515 , significant at the 1% level. This implies an immediate and persistent 0.515 percentage-point reduction in CIC/GDP relative to the no-PayNow counterfactual. Actual CIC/GDP lies below the counterfactual in 28 out of 29 post-intervention months. The coefficients of all ramp-intervention models are insignificant. Overall, the findings suggest that PayNow resulted in a one-time persistent reduction in aggregate cash demand, although the single-country experimental design warrants cautious causal interpretation.

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1. INTRODUCTION

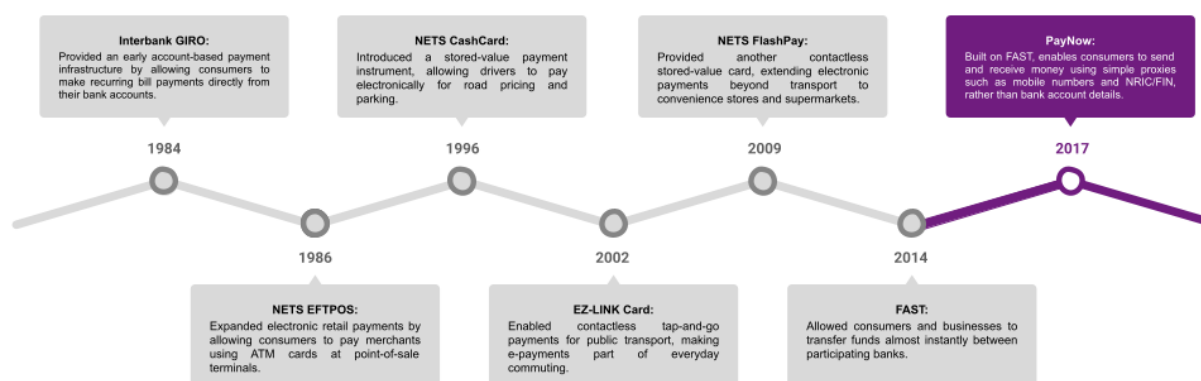
The rapid development of digital payment methods has revived one of the oldest questions in the history of economics: what determines the demand for cash? According to the traditional Baumol-Tobin inventory model (Baumol, 1952; Tobin, 1956), cash demand is determined by the trade-off between liquidity and opportunity cost. However, this model was developed in a context where cash was the main transaction instrument and therefore does not account for new payment technologies. As digital payment methods improve, the cost of accessing and transferring funds using such technology falls, hence incentivising individuals to conduct transactions with less cash. From this perspective, digital payments substitute cash because they increase the convenience of transferring funds and lower the cost of using non-cash payment methods (Organisation for Economic Co-operation and Development [OECD], 2018), which results in a decrease in overall cash demand (OECD, 2025).

However, existing empirical literature is more ambiguous than this theory implies. One strand finds evidence consistent with the substitution view, with Mercatanti and Li (2017) and Hwang and Wen (2024) documenting negative effects of digital payments on cash and money demand in Italy and China respectively. Yet a second strand documents the expansion of cash demand despite the diffusion of digital payments. For instance, Bech et al. (2018) show that cash remains widely used, reporting that across a sample of 50 economies, cash in circulation rose on average from around 7% of gross domestic product (GDP) in the early 2000s to around 9% of GDP by March 2018. The fragmentation of literature results thus motivates this present research of Singapore's PayNow system and its relationship with aggregate cash demand.

Singapore provides a useful setting for studying this question because its payments system combines advanced digital infrastructure with the continued relevance of cash (Government Technology Agency of Singapore, 2025; Chiang and Goh, 2026). As shown in Figure 1, the country's transition towards digital payments did not begin with PayNow. Earlier developments such as the General Interbank Recurring Order (Interbank GIRO) had already expanded non-cash payments in recurring and retail transactions. A more significant development happened with the introduction of Fast and Secure Transfers (FAST) in March 2014, which enabled near-real-time interbank transfers. However, FAST still required senders to know the recipient's bank name and account number, limiting its convenience for everyday peer-to-peer payments when compared to cash, which remained simple, anonymous, and widely accepted.

Figure 1: Timeline of Major Retail Payment Innovations in Singapore, 1984 – 2017

(Monetary Authority of Singapore, n.d.).



PayNow addressed this source of friction. Launched on 10 July 2017, PayNow allowed users of participating banks to transfer Singapore dollar funds almost instantly using the recipient's mobile number, national registration identity card (NRIC) number, or foreign identification number (FIN), rather than a bank account number. Built on FAST,

PayNow changed the practical convenience of account-to-account transfers overnight.^{1, 2}

To date, no research has examined whether the introduction of PayNow resulted in a measurable change in aggregate cash demand in Singapore. The closest papers to this study have instead examined related but different questions. Agarwal et al. (2019) used transaction-level data to investigate how the introduction of quick-response code (QR-code) payment technology through DBS PayLah!, a digital wallet launched by the Development Bank of Singapore, affected merchant outcomes. Similarly, Agarwal et al. (2025) examined the broader effects of the same QR-code technology on business creation, consumer spending, automated teller machine (ATM) withdrawals, and ATM closures. This paper thus contributes by providing the first direct empirical evidence on whether PayNow decreased aggregate cash demand in Singapore, and by further distinguishing between the adjustment mechanisms which PayNow's potential substitution effect could have operated through.

This research empirically examines the impact of PayNow on aggregate cash demand in Singapore. The central hypothesis is that the introduction of PayNow resulted in a substitution effect from cash towards digital payments, thereby lowering Singapore's

¹ It is important to note the distinction between PayNow and digital wallets such as DBS PayLah! and GrabPay. PayNow is a proxy-based system enabling direct bank transfers, whereas digital wallets are app-based platforms which may involve stored balances, rewards, and broader payment ecosystems.

² PayNow later expanded beyond peer-to-peer transfers through initiatives such as PayNow Corporate, PayNow QR, and cross-border linkages with systems including Thailand's PromptPay, India's UPI, and Malaysia's DuitNow. However, these subsequent developments are not modelled as separate interventions, as they were introduced in relatively quick succession. This leaves too few monthly observations between them to reliably identify distinct structural effects without imposing strong assumptions and increasing the risk of overfitting. As such, they are treated as extensions of the original PayNow infrastructure rather than independent interventions.

overall demand for physical currency. Using currency in circulation to gross domestic product (CIC/GDP) ratio as proxy for cash demand, this paper constructs a monthly CIC/GDP series for Singapore by temporally disaggregating quarterly nominal GDP into monthly GDP using the Chow-Lin method (Chow and Lin, 1971). It then estimates multiple seasonal autoregressive integrated moving average (SARIMA) intervention models around the introduction of PayNow, comparing a step specification with a ramp specification. After selecting the optimal baseline model based on statistical fit,³ coefficient validity, and residual diagnostic results, the paper employs an interrupted time-series (ITS) framework and constructs a no-PayNow counterfactual path for the CIC/GDP ratio. Robustness checks are also conducted through in-time placebo tests and a negative-control outcome (NCO) falsification test, both of which are reported in Appendix A.

This paper finds evidence consistent with a negative level shift in aggregate cash demand after PayNow. The selected baseline step-intervention model estimates a PayNow coefficient of -0.00515, statistically significant at the 1% level. This corresponds to a one-time 0.515 percentage-point decline in Singapore's CIC/GDP ratio almost immediately after PayNow's official launch. By contrast, the ramp specification is not statistically significant. Placebo tests using fake pre-PayNow intervention dates and the NCO falsification test do not produce statistically significant effects. These results support the paper's hypothesis. However, it is important to note

³ In this paper, the *baseline intervention model*, or simply *baseline model*, refers to the best candidate model selected for this paper's principal empirical analysis and subsequent counterfactual analysis. Depending on the results, the baseline model may take the form of either a baseline step-intervention model or a baseline ramp-intervention model.

A *step-intervention model* is a candidate model containing the PayNow step variable, while a *ramp-intervention model* is a candidate model containing the PayNow ramp variable (refer to Section 3.2).

that because this research's analysis is based on a single national time series with no untreated control group, the findings should be handled carefully and interpreted as suggestive causal evidence of PayNow's substitution relationship with cash demand rather than definitive proof of causality.

The rest of the paper proceeds as follows. Section 2 details data construction. Section 3 outlines the methodology. Section 4 presents results and counterfactual analysis. Section 5 concludes.

2. DATA

This paper utilises the following publicly available data to construct the empirical series used in the analysis: monthly end-of-period currency in circulation value, quarterly nominal GDP, monthly industrial production total index, monthly retail sales index excluding motor vehicles, monthly value of total domestic exports, monthly visitors' arrival, monthly unemployment rate, and monthly end-of-period Singapore Dollar against United States Dollar (SGD/USD) exchange rate.⁴

2.1. Outcome Variable

This paper uses a monthly adaptation of the standard CIC/GDP ratio as the outcome variable. To construct this series, quarterly nominal GDP is first temporally disaggregated into monthly estimates using the Chow-Lin method, before being

⁴ The replication code and dataset used in this paper are stored in the project repository at: https://github.com/nevinsey08/H3_HSSR_2026. The repository contains the materials used for temporal disaggregation, construction of the monthly CIC/GDP series, SARIMA intervention modelling, counterfactual analysis, and robustness tests.

annualised to ensure that the monthly CIC/GDP ratio remains comparable to the conventional annual measure. The construction of this monthly series is detailed in Sections 2.2 and 2.3.

Alternative proxies for cash demand are available but less suitable for this intervention analysis. For instance, ATM withdrawal value is a more direct proxy for transactional cash use (Khiaonarong and Humphrey, 2023), but the data available is semi-annual, making it too low-frequency for a precise ITS design around PayNow's launch. Temporally disaggregating ATM withdrawals into monthly observations would also be problematic, as it would impose smoothness and rely on indicators unlikely to reflect the same structural break (Chamberlin, 2010; Huang, 2012). Other ATM-based proxies, such as ATM withdrawals relative to GDP or total payment value, would therefore face the same limitation.

2.2. Temporal Disaggregation of Quarterly GDP

To construct the monthly CIC/GDP series, quarterly nominal GDP must first be converted into a monthly series. This paper does so by temporally disaggregating quarterly nominal GDP into monthly nominal GDP using the Chow-Lin method. Industrial production total index, retail sales index excluding motor vehicles, value of total domestic exports, visitors' arrival, and unemployment rate are used as the indicator series. Chow-Lin method is employed because the selected indicators are directly linked to Singapore's monthly economic activity and therefore are expected to exhibit a stable relationship with the nation's monthly GDP, while the remaining discrepancy is modelled as a stationary autoregressive process. As a robustness check, alternative temporal disaggregation approaches, including Fernández and

Litterman methods (Fernández, 1981; Litterman, 1983), were also implemented. These produced very similar monthly GDP estimates, as shown in Table 1 and Figure 2.

The aggregation constraint used in the Chow-Lin procedure is formalised as follows. Let g denote the vector of observed quarterly GDP values and z denote the unobserved vector of monthly GDP values. Since quarterly GDP is a flow variable, each quarterly value must equal the sum of the three corresponding monthly values. This aggregation relationship can be written as:

$$g = Cz \quad (1)$$

where C is the temporal aggregation matrix that maps monthly GDP into quarterly GDP taking the form:

$$C = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 & 1 & 1 \end{bmatrix} \quad (2)$$

The Chow-Lin method estimates the unobserved monthly series using the related monthly indicators. The high-frequency model is:

$$z = X\beta + u \quad (3)$$

where X denotes the matrix of monthly indicator variables, β is the coefficient vector, and u is the high-frequency disturbance term. The coefficient vector is estimated using the generalised least squares estimator:

$$\hat{\beta} = [X'C'(C\hat{\Omega}C')^{-1}CX]^{-1}X'C'(C\hat{\Omega}C')^{-1}g \quad (4)$$

where Ω is the covariance matrix capturing the assumed serial correlation structure of the monthly disturbance term.

Given $\hat{\beta}$, the Chow-Lin estimate of monthly GDP is:

$$\hat{z} = X\hat{\beta} + \hat{\Omega}C'(C\hat{\Omega}C')^{-1}(g - CX\hat{\beta}) \quad (5)$$

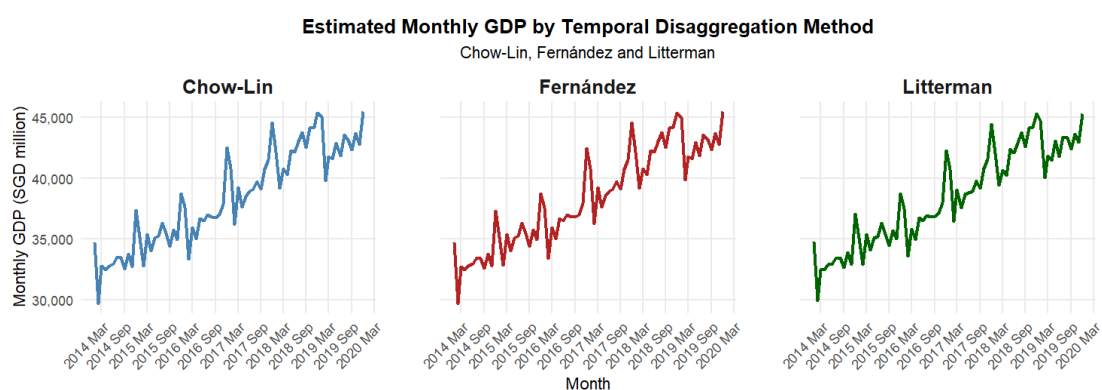
where, $\hat{z}$ denotes the estimated monthly nominal GDP.

Table 1: Summary Statistics for GDP Variables

Variable	N	Mean	S.D.	Start Date	End Date
Official quarterly GDP	24	115,231.4	11,515.2	Mar 2014	Dec 2019
Monthly GDP: Chow-Lin method	72	38,410.5	3,996.8	Jan 2014	Dec 2019
Monthly GDP: Fernández method	72	38,410.5	3,992.0	Jan 2014	Dec 2019
Monthly GDP: Litterman method	72	38,410.5	3,970.4	Jan 2014	Dec 2019

Note: All figures are in SGD million. The 2014 - 2019 sample restriction is discussed in Section 2.4.

Figure 2: Estimated Monthly Nominal GDP by Temporal Disaggregation Method.



2.3. Construction of Monthly CIC/GDP Ratio

A key measurement issue arises because CIC is a stock variable, while GDP is a flow variable. Directly dividing monthly CIC, written as CiC_t , by monthly GDP would therefore produce a ratio that is approximately 12 times larger than the conventional annual CIC/GDP ratio. To preserve the interpretation of the outcome variable so it matches the interpretation of the standard CIC/GDP ratio, monthly GDP is therefore annualised before constructing the ratio:

$$\text{Annualised monthly GDP}_t = 12 \times \hat{z}_t \quad (6)$$

The outcome variable monthly CIC/GDP ratio, denoted by y_t , is therefore:

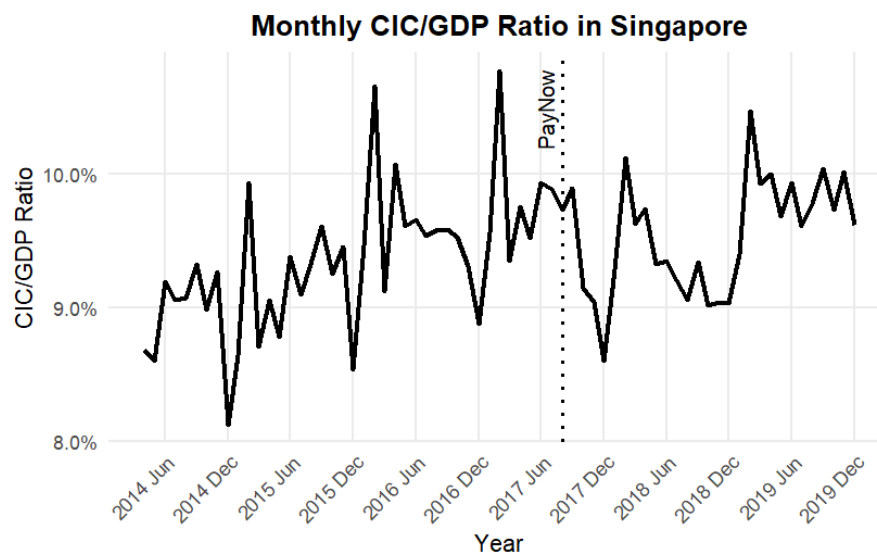
$$y_t = \frac{CiC_t}{12 \times \hat{z}_t} \quad (7)$$

Summary statistics for the constructed monthly CIC/GDP ratio are reported in Table 2, while Figure 3 plots the monthly CIC/GDP series for Singapore over the estimation window from April 2014 to December 2019.

Table 2: Summary Statistics for Monthly CIC/GDP Ratio

Period	N	Mean	S.D.	Start Date	End Date
Overall	69	0.0943	0.0050	Apr 2014	Dec 2019
Pre-PayNow	40	0.0935	0.0053	Apr 2014	Jul 2017
Post-PayNow	29	0.0954	0.0043	Aug 2017	Dec 2019

Note: The 2014–2019 sample restriction is discussed in Section 2.4.

Figure 3: Monthly CIC/GDP Ratio in Singapore, April 2014 – December 2019.

2.4. Sample Period

Although the constructed CIC/GDP series is available from March 1975 to December 2025, the main estimation window is restricted to April 2014 - December 2019. The lower bound is chosen as the first full month after the launch of FAST, Singapore's most recent institutional payments development preceding PayNow, to avoid confounding from earlier payment-system innovations. Meanwhile, the upper bound is selected to exclude COVID-19, which generated a large macroeconomic and behavioural shock affecting consumption and digital payment adoption.

The PayNow intervention is dated as August 2017, the first full month after its official launch on 10 July 2017. July 2017 is not used because PayNow was introduced partway through the month and was therefore not fully exposed to treatment. This same full-month convention also motivates setting the starting date for FAST as April 2014.

3. EMPIRICAL METHODOLOGY

The empirical design is a single-country interrupted time-series analysis, where the introduction of PayNow is treated as a distinct institutional intervention and the monthly CIC/GDP ratio is modelled.

3.1. Interrupted Time-Series Design

The ITS framework compares the observed post-intervention path of CIC/GDP with the counterfactual path that would have been expected in the absence of PayNow, conditional on the pre-existing structure of the series. The empirical design implemented in this paper closely follows the ARIMA-based ITS framework of Schaffer et al. (2021).

The intervention specification is defined formally below. Let y_t denote the monthly CIC/GDP ratio in month t , where $t = 1, 2, 3, \dots, T$. The estimation sample runs from April 2014 to December 2019, giving $T = 69$ monthly observations.

The general intervention model can be written as:

$$y_t = \beta I_t + \varepsilon_t \quad (8)$$

and:

$$I_t = \text{Step}_t \quad \text{or} \quad I_t = \text{Ramp}_t \quad (9)$$

where I_t is the PayNow intervention variable (elaborated on in Section 3.2), β measures the intervention effect, and ε_t is a serially correlated error component

following a SARIMA process which models the pre-existing dynamics of the CIC/GDP ratio.⁵

A conventional segmented-regression ITS design is not used in this research as CIC/GDP is a persistent monthly macroeconomic series and is expected to exhibit serial correlation (Glova and Hernandez, 2022), hence the assumption of independent errors will be violated, which results in misleading inference. A standard ARIMA model is also insufficient because it cannot explicitly model seasonality present in CIC/GDP (Cabrero et al., 2008; Consolvo and Lunsford, 2019).

⁵ Although y_t is defined as the level CIC/GDP ratio, equation (8) should be understood as the undifferenced intervention representation. Since the CIC/GDP ratio is expected to be persistent and non-stationary (see Figure 3), the model is not necessarily estimated in this undifferenced form once differencing is employed. In particular, Section 4 tests the level series and alternative transformations, including first, second, seasonal, and combined differences, to determine the appropriate differencing structure.

For example, if first non-seasonal differencing is applied, the intervention specification is instead written as:

$$\Delta y_t = \beta \Delta I_t + \Delta \varepsilon_t$$

where

$$\Delta y_t = y_t - y_{t-1}$$

If second non-seasonal differencing is applied, the corresponding form becomes:

$$\Delta^2 y_t = \beta \Delta^2 I_t + \Delta^2 \varepsilon_t$$

where

$$\Delta^2 y_t = y_t - 2y_{t-1} + y_{t-2}$$

If seasonal differencing is applied, the model becomes:

$$\Delta_{12} y_t = \beta \Delta_{12} I_t + \Delta_{12} \varepsilon_t$$

where

$$\Delta_{12} y_t = y_t - y_{t-12}$$

If both first non-seasonal and seasonal differencing are required, the differencing operators are applied jointly:

$$\Delta \Delta_{12} y_t = \beta \Delta \Delta_{12} I_t + \Delta \Delta_{12} \varepsilon_t$$

Thus, while equation (8) defines the intervention relationship in levels, the actual SARIMA estimation may be conducted using the corresponding differenced form of the intervention-adjusted series, depending on the stationarity properties of y_t .

3.2. PayNow Intervention Variable

Two alternative PayNow intervention specifications are estimated: a step specification and a ramp specification. These correspond to two different economic hypotheses about how PayNow may affect cash demand.

The step variable tests whether PayNow produced an immediate level shift in aggregate cash demand. Economically, this corresponds to the theory that PayNow lowered the stock of CIC relative to GDP by reducing the transactional need for cash immediately after launch. It is given as:

$$Step_t = \mathbb{1}(t \geq \tau) \quad (10)$$

where τ denote the PayNow intervention month of August 2017.

The ramp variable tests whether PayNow produced a gradual post-intervention trend change. Economically, this corresponds to the theory that the substitution from cash to PayNow increased progressively over time as user adoption, merchant acceptance, and familiarity with digital transfers expanded. It is defined as:

$$Ramp_t = (t - \tau + 1) \times \mathbb{1}(t \geq \tau) \quad (11)$$

The step and ramp specifications are estimated separately in the baseline analysis. In a short monthly sample (69 observations), estimating both variables jointly can result in collinearity as both variables change only after the same intervention date. Separate estimation thus provides a cleaner and more accurate comparison.

3.3. SARIMA Error Structure

The serially correlated component ε_t is modelled using a SARIMA process. Let B denote the backshift operator, such that:

$$By_t = y_{t-1} \quad (12)$$

and:

$$B^{12}y_t = y_{t-12} \quad (13)$$

where 12 is the seasonal period for monthly data.

The general SARIMA error process with drift is written as:

$$\Phi(B^{12})\varphi(B)(1-B)^d(1-B^{12})^D\varepsilon_t = \mu + \theta(B^{12})\theta(B)u_t \quad (14)$$

where μ is the drift term, $\varphi(B)$ is the non-seasonal autoregressive (AR) polynomial of order p , $\theta(B)$ is the non-seasonal moving-average (MA) polynomial of order q , $\Phi(B^{12})$ is the seasonal autoregressive polynomial of order P , $\theta(B^{12})$ is the seasonal moving-average polynomial of order Q , d is the order of non-seasonal differencing, D is the order of seasonal differencing, and u_t is a white-noise innovation.

3.4. Counterfactual Model

After selecting the optimal baseline intervention model, the paper constructs a no-PayNow counterfactual path for monthly CIC/GDP ratio denoted as y_t^0 .

The fitted value from the baseline intervention model can be written as:

$$\hat{y}_t = \hat{\beta}I_t + \hat{\varepsilon}_t \quad (15)$$

where $\hat{y}_t$ is the fitted value and $\hat{\varepsilon}_t$ includes the fitted SARIMA component.

The no-PayNow counterfactual is constructed as:

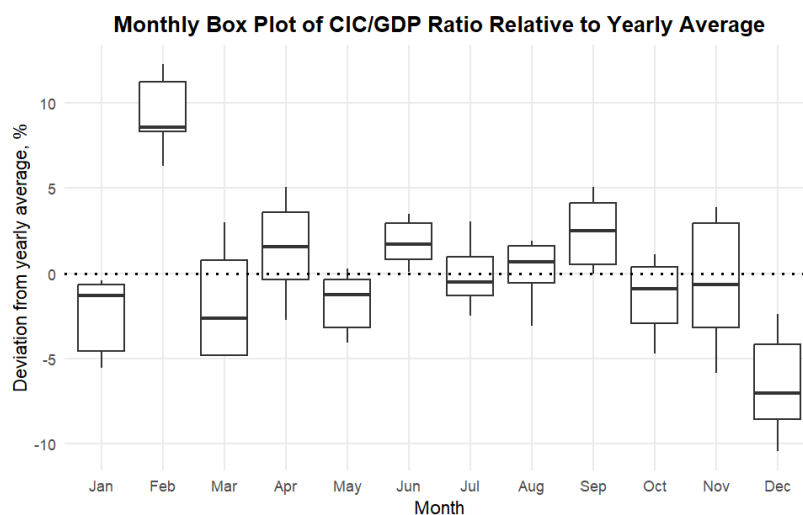
$$\hat{y}_t^0 = \hat{\varepsilon}_t \quad (16)$$

where $\hat{y}_t^0$ is the model-implied CIC/GDP ratio in the absence of PayNow.

3.5. Assumptions

First, the timing of PayNow must be plausibly exogenous to short-run fluctuations in the CIC/GDP ratio. If the intervention month coincides with a month that has a strong recurring seasonal pattern, the SARIMA intervention model will mistakenly attribute a predictable seasonal movement to PayNow. Figure 4 shows that some months exhibit much stronger seasonal deviations than others. For example, CIC/GDP is consistently above the yearly average in February which can be attributed to higher cash demand due to Chinese New Year cash-giving practices in Singapore. If PayNow had been introduced in February, any observed increase in CIC/GDP would reflect this regular seasonal pattern rather than the causal effect of PayNow. By contrast, August does not display an unusually large recurring deviation from the yearly average. This reduces the concern that the estimated PayNow effect is simply capturing a predictable seasonal fluctuation rather than an intervention effect.

Figure 4: Monthly Seasonality in CIC/GDP Ratio Relative to Yearly Average.



Second, in the absence of PayNow, monthly CIC/GDP ratio would have continued to evolve according to its pre-existing stochastic trend and seasonal structure. The counterfactual assumption can be written as:

$$y_t^0 \approx \hat{y}_t^0 \quad \text{when } t \geq \tau \quad (17)$$

where y_t^0 denotes the actual unobserved no-PayNow path of CIC/GDP.

Third, there should be no other contemporaneous structural changes prior to and during August 2017 that independently caused breaks in aggregate cash demand. The paper mitigates this concern by restricting the sample from April 2014 to December 2019, as explained in Section 2.4.

4. MAIN RESULTS

Table 3: Stationarity Tests for Monthly CIC/GDP Ratio

Series	N	ADF statistic	KPSS statistic
Level	69	-2.8036 (p = 0.249)	0.7209 (p = 0.012)
First difference	68	-5.0940 (p < 0.010)	0.0403 (p > 0.100)
Second difference	67	-6.3175 (p < 0.010)	0.0312 (p > 0.100)
Seasonal difference	57	-1.7278 (p = 0.684)	0.2456 (p > 0.100)
First + seasonal difference	56	-4.0366 (p = 0.014)	0.0944 (p > 0.100)
Second + seasonal difference	55	-5.2147 (p < 0.010)	0.0636 (p > 0.100)

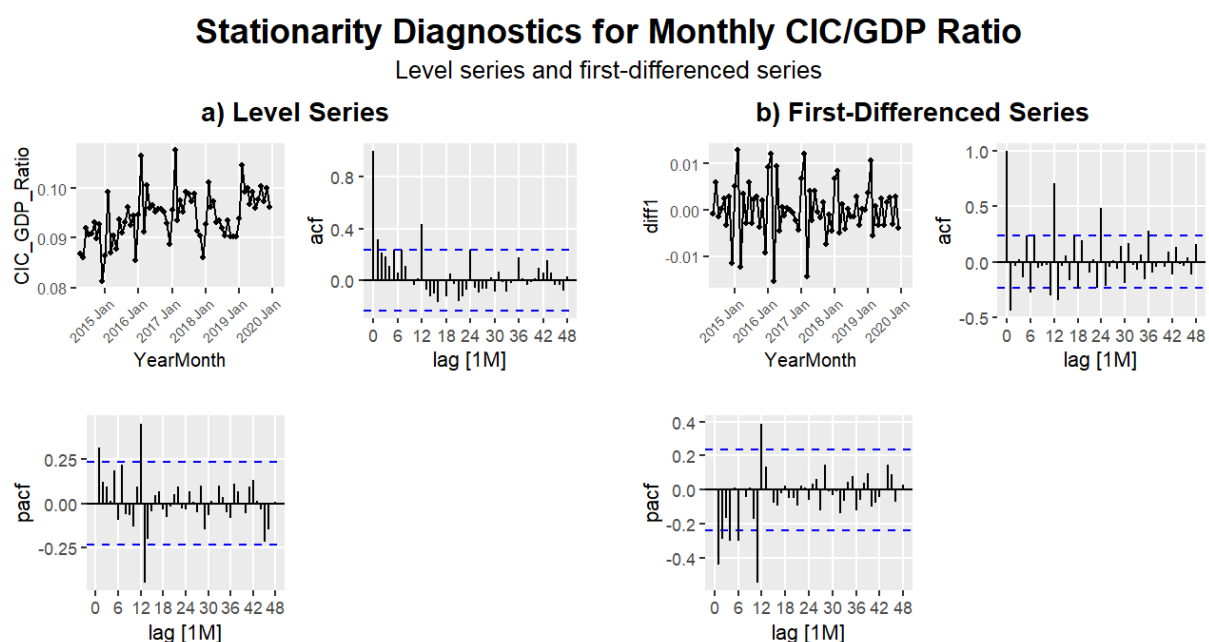
Note: p-values are presented in parentheses.

Table 3 reports the Augmented Dickey-Fuller (ADF) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests for the level CIC/GDP series and alternative

transformations. The level series is non-stationary, with the ADF test failing to reject the null hypothesis of a unit root with a test statistic of -2.8036 and p-value of 0.249 . The KPSS test, which takes stationarity as the null hypothesis, also rejects stationarity, with a statistic of 0.7209 and p-value of 0.012 .

First differencing resolves this problem. The first-differenced series has an ADF statistic of -5.0940 and p-value < 0.010 , rejecting the null hypothesis of a unit root. The KPSS statistic falls sharply to 0.0403 , with p-value > 0.100 , which means that stationarity can no longer be rejected. This indicates that one non-seasonal difference is sufficient to remove the pre-existing trend in the monthly CIC/GDP series. Since first differencing alone already produces a stationary series, no further differencing transformation is utilised to avoid over-differencing the data. Therefore, all candidate SARIMA models are estimated with non-seasonal differencing $d = 1$ and seasonal differencing $D = 0$.

Figure 5: Time Series, ACF, and PACF Plots of Level and First-Differenced Monthly CIC/GDP Ratio.



From Figure 5, in the level series, the ACF decays slowly and shows visible spikes around annual lags, indicating persistence and seasonal dependence. After first differencing, the series becomes more stable around zero, but the ACF and PACF still show notable short-run and seasonal autocorrelation, especially around lag 1 and lag 12. This suggests that additional non-seasonal and seasonal ARMA terms are needed to model the remaining dependence structure. Accordingly, manual step and ramp SARIMA intervention models are estimated. The models are written as:

$$\text{step/ramp}_{p1q_P0Q}$$

with the following equation:

$$\Phi(B^{12})\varphi(B)(1-B)\varepsilon_t = \mu + \theta(B^{12})\theta(B)u_t \quad (18)$$

Since the ACF and PACF do not identify a single unambiguous specification, multiple theoretically plausible candidate intervention models are estimated and compared as summarised in Table 4.

Table 4: Candidate SARIMA Intervention Models.

Model Name	ARIMA order	Seasonal order
step/ramp_auto	Auto-selected with d = 1	Auto-selected with D = 0
step/ramp_111_101	(1,1,1)	(1,0,1)[12]
step/ramp_011_101	(0,1,1)	(1,0,1)[12]
step/ramp_011_100	(0,1,1)	(1,0,0)[12]
step/ramp_110_101	(1,1,0)	(1,0,1)[12]
step/ramp_111_100	(1,1,1)	(1,0,0)[12]
step/ramp_110_100	(1,1,0)	(1,0,0)[12]

Model Name	ARIMA order	Seasonal order
step/ramp_111_001	(1,1,1)	(0,0,1)[12]
step/ramp_011_001	(0,1,1)	(0,0,1)[12]
step/ramp_110_001	(1,1,0)	(0,0,1)[12]

In addition, auto-selected SARIMA models are estimated separately for the step and ramp specifications, while still imposing $d = 1$ and $D = 0$. These auto-selected specifications are included for reference and comparison.

4.1. Step-Intervention Results

Table 5: Coefficient Estimates for Step SARIMA Models

Model	ARIMA order	Seasonal order	PayNow step	Drift Constant	AR(1)	MA(1)	Seasonal AR(1)	Seasonal MA(1)
step_auto	Auto-selected with $d = 1$	Auto-selected with $D = 0$	-0.00104 (0.00185)	0.00032 (0.00064)		-0.47094*** (0.12831)	0.83250*** (0.05684)	
step_111_101	(1,1,1)	(1,0,1)[12]	-0.00515*** (0.00182)	0.00024 (0.00015)	0.48636*** (0.12846)	-1.00000*** (0.01477)	0.94246*** (0.04232)	-0.39790** (0.18226)
step_011_101	(0,1,1)	(1,0,1)[12]	-0.00189 (0.00198)	0.00033 (0.00067)		-0.47965*** (0.12897)	0.92607*** (0.04748)	-0.31473* (0.16945)
step_011_100	(0,1,1)	(1,0,0)[12]	-0.00104 (0.00185)	0.00032 (0.00064)		-0.47094*** (0.12831)	0.83250*** (0.05684)	
step_110_101	(1,1,0)	(1,0,1)[12]	-0.00133 (0.00207)	0.00036 (0.00100)	-0.37124*** (0.11302)		0.92860*** (0.04498)	-0.28791* (0.16851)
step_111_100	(1,1,1)	(1,0,0)[12]	-0.00165 (0.00226)	0.00031 (0.00048)	0.26651 (0.38579)	-0.71029** (0.32530)	0.82639*** (0.05909)	
step_110_100	(1,1,0)	(1,0,0)[12]	-0.00076 (0.00195)	0.00034 (0.00093)	-0.37335*** (0.11355)		0.84907*** (0.05090)	
step_111_001	(1,1,1)	(0,0,1)[12]	-0.00298 (0.00225)	0.00019 (0.00016)	0.29155** (0.12744)	-1.00000*** (0.04072)		0.61147*** (0.13524)
step_011_001	(0,1,1)	(0,0,1)[12]	-0.00073 (0.00227)	0.00019 (0.00022)		-0.74535*** (0.12288)		0.63942*** (0.13317)

Model	ARIMA order	Seasonal order	PayNow step	Drift Constant	AR(1)	MA(1)	Seasonal AR(1)	Seasonal MA(1)
step_110_001	(1,1,0)	(0,0,1)[12]	0.00056 (0.00271)	0.00013 (0.00053)	-0.41715*** (0.11065)			0.71096*** (0.13929)

*Note: Estimated coefficients are reported with standard errors in parentheses. ***, **, and * denote statistical significance based on p-values at the 1% ($p < 0.01$), 5% ($p < 0.05$), and 10% ($p < 0.10$) levels, respectively.*

Table 5 reports the coefficient estimates for the step-intervention SARIMA models. Across the step specifications, the estimated PayNow coefficient is generally negative. This is consistent with the substitution hypothesis. However, the statistical strength of this result differs substantially across models. The strongest evidence for an immediate one-time PayNow substitution effect comes from step_111_101. In this specification, the PayNow step coefficient is -0.00515 , with a standard error of 0.00182 . The coefficient is statistically significant at the 1% level. The other step models produce weaker evidence of a substitution effect. Most of their PayNow coefficients are negative, but all of them are not statistically significant even at the 10% significance level.

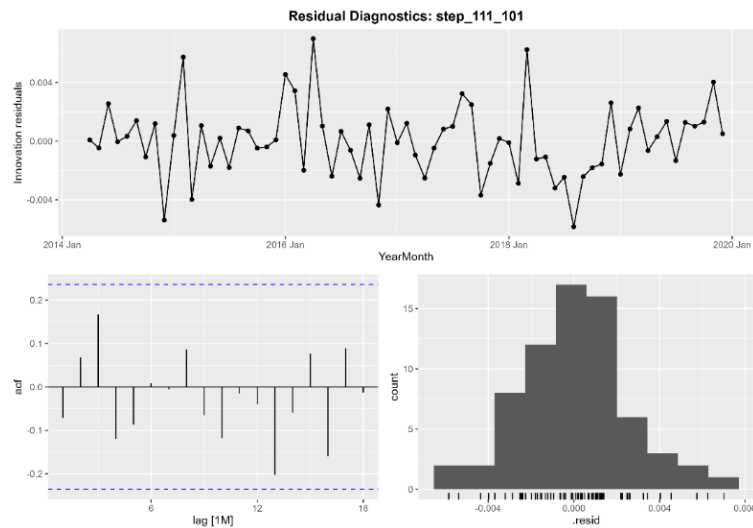
Table 6: Ljung-Box Residual Diagnostic Test for Step SARIMA Models

Model	ARIMA order	Seasonal order	Ljung-Box statistic
step_auto	Auto-selected, $d = 1$	Auto-selected, $D = 0$	10.3660 ($p = 0.498$)
step_111_101	(1,1,1)	(1,0,1)[12]	10.2717 ($p = 0.329$)
step_011_101	(0,1,1)	(1,0,1)[12]	9.2111 ($p = 0.512$)
step_011_100	(0,1,1)	(1,0,0)[12]	10.3660 ($p = 0.498$)
step_110_101	(1,1,0)	(1,0,1)[12]	11.4223 ($p = 0.326$)
step_111_100	(1,1,1)	(1,0,0)[12]	8.2703 ($p = 0.602$)
step_110_100	(1,1,0)	(1,0,0)[12]	11.3460 ($p = 0.415$)
step_111_001	(1,1,1)	(0,0,1)[12]	15.4784 ($p = 0.116$)

Model	ARIMA order	Seasonal order	Ljung-Box statistic
step_011_001	(0,1,1)	(0,0,1)[12]	12.1442 (p = 0.353)
step_110_001	(1,1,0)	(0,0,1)[12]	18.0344 (p = 0.081)

Note: *p*-values are presented in parentheses.

Figure 6: Visual Residual Diagnostic Tests for step_111_101 Model.⁶



From Table 6 and Figure 6, the residual diagnostic results support the admissibility of the step models. In this paper, the Ljung-Box test lag is set as $\min(2s, \frac{T}{5})$, where $s = 12$ is the seasonal period for monthly data and $T = 69$ is the number of observations (Hyndman, 2014). Across all step specifications, the null hypothesis of no residual autocorrelation cannot be rejected. The visual diagnostics are also generally acceptable,⁷ although some weaker specifications display larger residual dispersion and isolated autocorrelation features. These issues do not overturn the Ljung-Box results but provide further justification for not selecting those models as the baseline.

⁶ Visual residual diagnostic plots for the remaining step-intervention candidate models are reported in Appendix B owing to space constraints.

⁷ In this paper, visual residual diagnostics is conducted through the analysis of residual time plots, residual ACF plots, and residual histogram with rug plots, as displayed in Figure 7.

Table 7: Model Selection Metrics for Step SARIMA Models

Model	ARIMA order	Seasonal order	σ^2	Log likelihood	AIC	AICc	BIC
step_auto	Auto-selected with d = 1	Auto-selected with D = 0	7.758e-06	298.434	-586.867	-585.899	-575.769
step_111_101	(1,1,1)	(1,0,1)[12]	6.789e-06	302.543	-591.086	-589.220	-575.550
step_011_101	(0,1,1)	(1,0,1)[12]	7.255e-06	300.141	-588.281	-586.904	-574.964
step_011_100	(0,1,1)	(1,0,0)[12]	7.758e-06	298.434	-586.867	-585.899	-575.769
step_110_101	(1,1,0)	(1,0,1)[12]	7.436e-06	298.866	-585.732	-584.355	-572.415
step_111_100	(1,1,1)	(1,0,0)[12]	7.867e-06	298.652	-585.303	-583.926	-571.986
step_110_100	(1,1,0)	(1,0,0)[12]	7.878e-06	297.391	-584.782	-583.814	-573.684
step_111_001	(1,1,1)	(0,0,1)[12]	1.248e-05	285.822	-559.644	-558.267	-546.327
step_011_001	(0,1,1)	(0,0,1)[12]	1.330e-05	283.793	-557.585	-556.617	-546.488
step_110_001	(1,1,0)	(0,0,1)[12]	1.488e-05	279.242	-548.485	-547.517	-537.387

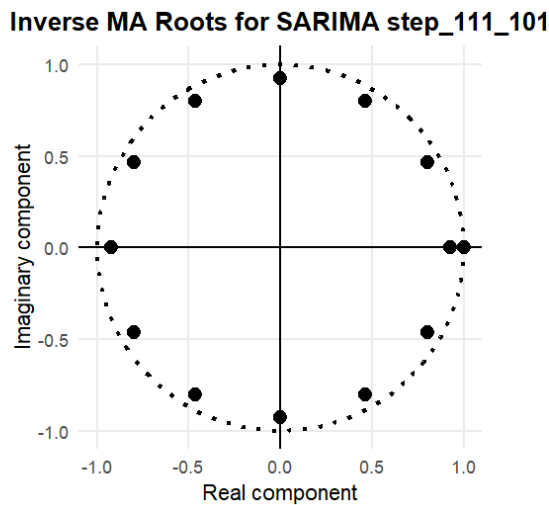
*Note: Estimated coefficients are reported with standard errors in parentheses. ***, **, and * denote statistical significance based on p-values at the 1% ($p < 0.01$), 5% ($p < 0.05$), and 10% ($p < 0.10$) levels, respectively.*

The model-selection metrics in Table 7 provide the strongest case for choosing step_111_101 as the baseline intervention model. This model has the lowest residual variance, AIC, AICc, and highest log likelihood, thus making it the best-performing step model across the full set of criteria. Although its BIC ranks lower than other models, this does not overturn the selection, since BIC imposes a stronger penalty on parameter count and thus may favour excessive parsimony in a short but dynamically complex monthly series. Therefore, step_111_101 is selected as the preferred baseline step-intervention model, which will be used to construct a no-PayNow path in Section 4.3.

Table 8: Inverse MA Root Diagnostics for SARIMA step_111_101

Root number	Root modulus	Inverse root modulus
1	1.07982	0.92608
2	1.07982	0.92608
3	1.07982	0.92608
4	1.00000	1.00000
5	1.07982	0.92608
6	1.07982	0.92608
7	1.07982	0.92608
8	1.07982	0.92608
9	1.07982	0.92608
10	1.07982	0.92608
11	1.07982	0.92608
12	1.07982	0.92608
13	1.07982	0.92608

Note: An MA process is invertible when all inverse-root moduli are less than one.

Figure 7: Inverse Moving-Average Roots of step_111_101.

However, it is important to note that the estimated non-seasonal MA coefficient in step_111_101 is approximately -1.0000 . This places the model very close to the invertibility boundary, which raises concerns about its numerical stability and the reliability to construct a counterfactual. To address this, the roots of the full MA polynomial were explicitly examined. With reference to Table 8 and Figure 7, the inverse-root diagnostic indicates that the model is right on the boundary of invertibility, with all inverse roots lying inside, but on the edge, of the unit circle. This implies that the model is numerically near-boundary.

This caveat is important, but it does not invalidate the model choice. The selected model still has the best overall fit, passes the Ljung-Box test, and shows acceptable visual residual diagnostic results. Hence, step_111_101 remains as the preferred step-specification baseline model, while the near-boundary MA coefficient is treated as a limitation to be kept in mind when interpreting the results.

4.2. Ramp-Intervention Results

Table 9: Coefficient Estimates for Ramp SARIMA Models

Model	ARIMA order	Seasonal order	PayNow ramp	Drift Constant	AR(1)	MA(1)	Seasonal AR(1)	Seasonal MA(1)
ramp_auto	Auto-selected with d = 1	Auto-selected with D = 0	-0.00057 (0.00047)	0.00059 (0.00069)		-0.46790*** (0.11971)	0.84018*** (0.05476)	
ramp_011_100	(0,1,1)	(1,0,0)[12]	-0.00057 (0.00047)	0.00059 (0.00069)		-0.46790*** (0.11971)	0.84018*** (0.05476)	
ramp_011_101	(0,1,1)	(1,0,1)[12]	-0.00040 (0.00045)	0.00052 (0.00074)		-0.45831*** (0.11839)	0.91368*** (0.05286)	-0.25105 (0.17429)
ramp_110_100	(1,1,0)	(1,0,0)[12]	-0.00074 (0.00060)	0.00069 (0.00098)	-0.38606*** (0.11263)		0.85469*** (0.04963)	
ramp_110_101	(1,1,0)	(1,0,1)[12]	-0.00054 (0.00058)	0.00062 (0.00103)	-0.37948*** (0.11261)		0.91989*** (0.04843)	-0.24191 (0.17175)
ramp_111_100	(1,1,1)	(1,0,0)[12]	-0.00054 (0.00050)	0.00057 (0.00068)	0.05250 (0.34979)	-0.51478 (0.32495)	0.83883*** (0.05577)	
ramp_111_101	(1,1,1)	(1,0,1)[12]	-0.00040 (0.00046)	0.00052 (0.00074)	0.01333 (0.25189)	-0.47028* (0.24737)	0.91347*** (0.05014)	-0.25088 (0.16873)
ramp_011_001	(0,1,1)	(0,0,1)[12]	-0.00029 (0.00038)	0.00030 (0.00028)		-0.72213*** (0.11030)		0.67165*** (0.13188)
ramp_111_001	(1,1,1)	(0,0,1)[12]	-0.00016 (0.00034)	0.00024 (0.00026)	0.22301 (0.24594)	-0.85929*** (0.18084)		0.67905*** (0.13587)
ramp_110_001	(1,1,0)	(0,0,1)[12]	-0.00070 (0.00081)	0.00042 (0.00062)	-0.42155*** (0.11016)			0.72570*** (0.14680)

*Note: Estimated coefficients are reported with standard errors in parentheses. ***, **, and * denote statistical significance based on p-values at the 1% ($p < 0.01$), 5% ($p < 0.05$), and 10% ($p < 0.10$) levels, respectively.*

Table 9 reports the coefficient estimates for the ramp-intervention SARIMA models. The ramp coefficients are negative across the candidate models, which are consistent with the substitution hypothesis. However, none of the ramp coefficients is statistically significant at conventional levels. This implies that there is no strong evidence which

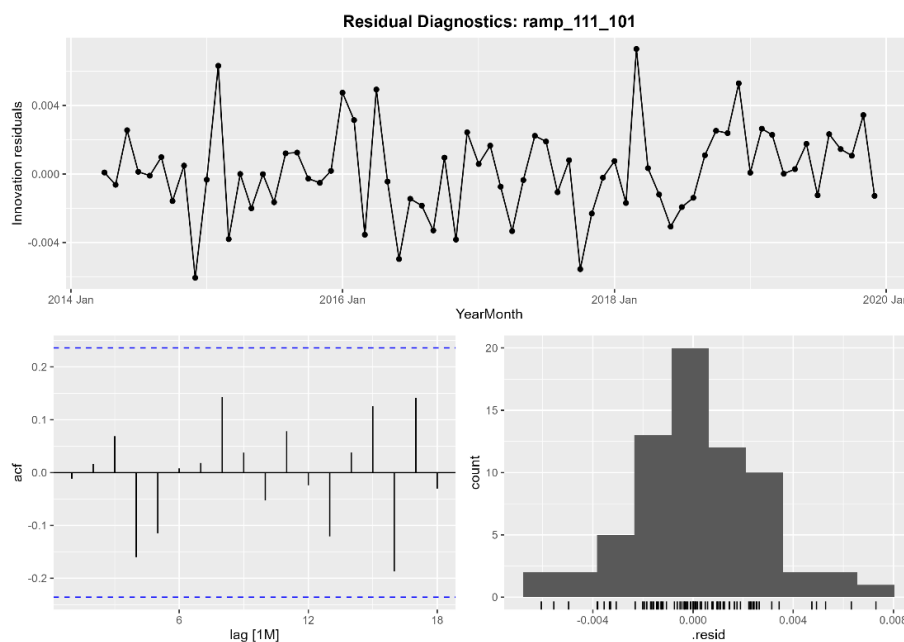
suggests PayNow generated a gradual increasing decline in cash demand after its introduction.

Table 10: Ljung-Box Residual Diagnostic Test for Ramp SARIMA Models

Model	ARIMA order	Seasonal order	Ljung-Box statistic
ramp_auto	Auto-selected, d = 1	Auto-selected, D = 0	8.2147 (p = 0.694)
ramp_011_100	(0,1,1)	(1,0,0)[12]	8.2147 (p = 0.694)
ramp_011_101	(0,1,1)	(1,0,1)[12]	7.2338 (p = 0.703)
ramp_110_100	(1,1,0)	(1,0,0)[12]	10.0350 (p = 0.527)
ramp_110_101	(1,1,0)	(1,0,1)[12]	9.8792 (p = 0.451)
ramp_111_100	(1,1,1)	(1,0,0)[12]	7.8101 (p = 0.647)
ramp_111_101	(1,1,1)	(1,0,1)[12]	7.1818 (p = 0.618)
ramp_011_001	(0,1,1)	(0,0,1)[12]	11.2137 (p = 0.426)
ramp_111_001	(1,1,1)	(0,0,1)[12]	12.0259 (p = 0.283)
ramp_110_001	(1,1,0)	(0,0,1)[12]	17.9095 (p = 0.084)

Note: p-values are presented in paratheses.

Figure 8: Visual Residual Diagnostic Tests for ramp_111_101 Model.⁸



⁸ Visual residual diagnostic plots for the remaining ramp-intervention candidate models are reported in Appendix B owing to space constraints.

Table 10 and Figure 8 indicate that the residual diagnostic results for the ramp models are satisfactory. Using Ljung-Box tests, all ramp models show no strong evidence of residual autocorrelation. The visual residual plots of all models are also broadly acceptable. Nevertheless, no ramp specification is selected as the baseline intervention model because the estimated PayNow ramp coefficient is statistically insignificant across all ramp-intervention models.

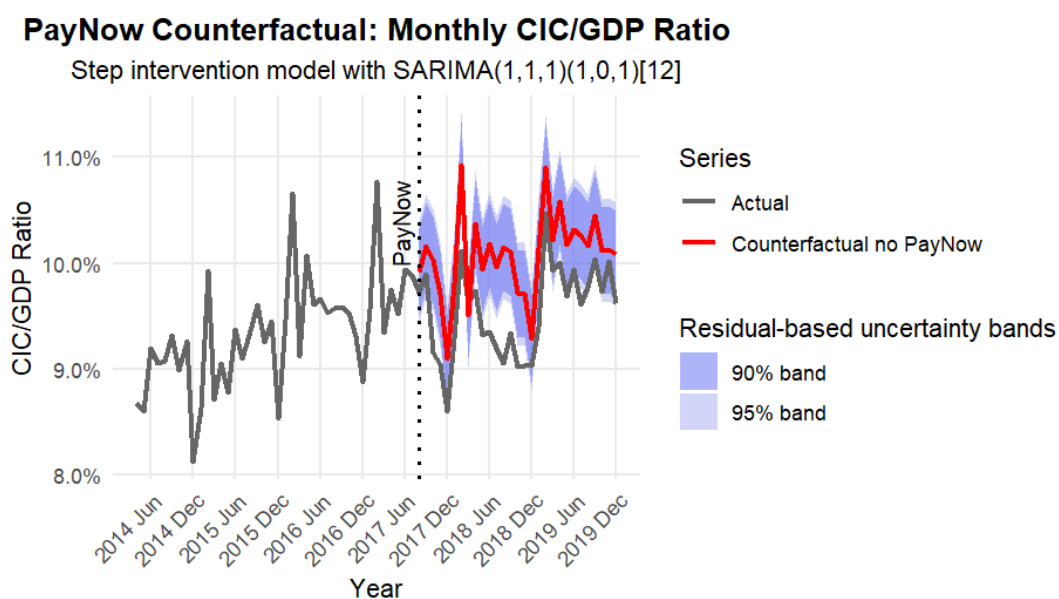
4.3. Counterfactual Analysis

Table 11: SARIMA Step Counterfactual Results

PayNow step coefficient	Post-PayNow months	Months below counterfactual	Months below 90% band	Months below 95% band
-0.00515*** (0.00182)	29	28	20	16

Note: Estimated coefficients are reported with standard errors in parentheses. ***, **, and * denote statistical significance based on p -values at the 1% ($p < 0.01$), 5% ($p < 0.05$), and 10% ($p < 0.10$) levels, respectively.

Figure 9: Actual and No-PayNow Counterfactual Monthly CIC/GDP Ratio, April 2014 - December 2019.



The selected step_111_101 model is used to construct a no-PayNow counterfactual path for the monthly CIC/GDP ratio, as defined in equation (16). As reported in Table 11, the estimated PayNow step coefficient is -0.00515 , equivalent to an immediate one-time reduction of 0.515 percentage points in CIC/GDP ratio relative to the model-implied counterfactual. This result is consistent with the paper's hypothesis that PayNow reduced aggregate cash demand. Furthermore, Figure 9 shows that the actual CIC/GDP series lies persistently below the no-PayNow counterfactual after the intervention. Across the 29 post-PayNow months, actual CIC/GDP is below the counterfactual in 28 months, below the lower 90% residual-based band in 20 months, and below the lower 95% band in 16 months. This indicates that the divergence is persistent across the post-PayNow window, rather than being driven by a single outlier or by stochastic and seasonal variation alone.

This counterfactual interpretation relies on the strong assumption that in the absence of PayNow, monthly CIC/GDP would have continued to evolve according to its pre-existing stochastic trend and seasonal structure, as discussed in Section 3.5. Since the actual no-PayNow path is unobserved, this assumption cannot be directly tested. The counterfactual results should therefore be interpreted cautiously as suggestive evidence of a PayNow-related reduction in cash demand, rather than as strong and direct proof of causality.

Nevertheless, the robustness checks reported in Appendix A strengthen the counterfactual results. First, in-time placebo tests are conducted to re-estimate the baseline model using fake intervention dates before PayNow's launch. If the estimated effect were driven by a pre-existing decline in CIC/GDP, these placebo dates should

also produce statistically significant step effects. Instead, the results show that fake pre-PayNow intervention dates do not produce statistically significant step effects, reducing the concern that the counterfactual gap is driven by an already-existing downward shift in CIC/GDP before PayNow. Second, the NCO falsification test applies the same intervention structure to the SGD/USD exchange rate, an outcome that should not plausibly respond to PayNow. The statistically insignificant PayNow coefficient reported in this test reduces the concern that the August 2017 break reflects a broader macro-financial shock or a mechanical tendency of the SARIMA specification to identify a break around the intervention date.

The residual-based uncertainty bands should also be interpreted cautiously. They are constructed by adding multiples of the estimated residual standard deviation to the counterfactual path, and therefore represent descriptive uncertainty ranges rather than formal confidence or prediction intervals. Because the counterfactual is obtained from a full-sample fitted intervention model by removing the estimated PayNow component, formal intervals would need to account for parameter uncertainty, model-selection uncertainty, innovation variability, and uncertainty from the temporal disaggregation of GDP. Since the bands capture only residual variation, they are used solely as approximate descriptive guides.

4.4. Discussion

The contrast between the step and ramp results is informative because it sheds light on the mechanism through which PayNow may have affected cash demand. A step effect implies an immediate one-time downward shift in the level of CIC/GDP after the intervention, with subsequent stochastic and seasonal movements continuing with this

lower level. A ramp effect, by contrast, implies a gradual constant decrease of CIC/GDP, causing the gap between actual and counterfactual cash demand to widen progressively over time. Overall, the results indicate that PayNow's substitution effect operated primarily through an immediate and persistent downward level shift, rather than through a progressively widening post-intervention decline.

Such a rapid adjustment is plausible in Singapore's institutional and technological context. PayNow was introduced into an economy with widespread internet access, high smartphone penetration (Infocomm Media Development Authority Singapore, 2023), and substantial familiarity with online and mobile banking (Breeze, 2017). FAST had also introduced to households and firms the underlying infrastructure for real-time interbank transfers before PayNow's launch. PayNow therefore did not require users to adopt an entirely unfamiliar technology, which enabled such a rapid substitution effect. Moreover, the resulting reduction in transaction costs and behavioural frictions in using PayNow's FAST-based transaction system may have been large enough for households and firms to revise their desired cash holdings within a relatively short period.

The absence of a statistically significant ramp effect is also plausible. PayNow mainly displaces a fixed set of routine cash uses, including peer-to-peer transfers and recurring day-to-day payments (PricewaterhouseCoopers, 2026). The value of such transactions is constrained by households' regular expenditure and transfer needs. Once these transactions shift from cash to PayNow, further growth in usage slows down significantly. The effect on CIC/GDP may therefore not be a linear decrease, with the largest adjustment occurring when PayNow first becomes a sufficiently convenient

substitute for cash followed by exponentially diminishing marginal effects. Moreover, continued expansion in PayNow's user base need not imply a progressively declining level of aggregate cash demand. New users may adopt PayNow without reducing their currency holdings, with the constant cash balances reflecting a change in the function of cash towards precautionary (Pietrucha and Maciejewski, 2020), store-of-value (Faella and Zamora-Pérez, 2025), and fallback purposes (Alvarez et al., 2026).

5. CONCLUSION

In conclusion, this paper examines whether the introduction of PayNow is associated with a decrease in aggregate cash demand in Singapore. It constructs a monthly CIC/GDP ratio using monthly CIC data and monthly GDP estimates derived by temporally disaggregating quarterly nominal GDP. Using this series, the paper estimates SARIMA-ITS models, treating August 2017 as the PayNow intervention month.

The main finding is that PayNow is associated with a statistically significant decline in aggregate cash demand. The selected step_111_101 model estimates a PayNow coefficient of -0.00515 , equivalent to a 0.515 percentage-point fall in CIC/GDP. The counterfactual analysis further shows that actual CIC/GDP lies below the no-PayNow counterfactual in 28 out of 29 post-intervention months. In-time placebo tests and the NCO falsification test also do not produce statistically significant effects, hence reducing the likelihood that the baseline-intervention model result reflects a pre-existing trend, a mechanical SARIMA feature, or a broader macro-financial shock.

Nevertheless, the findings should be interpreted with caution given the various assumptions and limitations discussed throughout this paper. The evidence is therefore best understood as suggestive rather than definitive causal evidence. Ultimately, this paper finds that PayNow contributed to a meaningful reduction in Singapore's aggregate cash demand, consistent with the interpretation that households and firms substituted the use of cash with PayNow almost immediately after its introduction, with Singapore's economy having adjusted to a lower equilibrium level of cash usage.

(Word count: 4500 — excluding footnotes, equations, figure titles, figures, table titles, and tables)

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7. APPENDIX A

In-Time Placebo Tests

One of the biggest threats to identification of PayNow's effect is that the estimated step-coefficient may reflect either an unaccounted for pre-existing decline in CIC/GDP or a mechanical tendency of the SARIMA baseline model to attribute significance to any step indicator included in the model specification. If either of these were true, placebo interventions assigned to dates before PayNow's introduction should also generate statistically significant coefficients.

To assess this, the selected baseline step-intervention model is re-estimated using fake intervention dates prior to PayNow. Each placebo model is estimated exclusively on the pre-PayNow sample from April 2014 to July 2017, thereby preventing the true intervention from influencing the placebo estimates.

The chosen placebo dates are November 2016, December 2016, January 2017, February 2017, March 2017, April 2017, and May 2017. However, the specifications using November 2016 and January 2017 could not be estimated successfully because the parameter-rich seasonal SARIMA model failed to converge reliably within the short pre-intervention sample. These two models are hence omitted in the results.

Table 12: In-Time Placebo Step Coefficient Estimates

Placebo date	ARIMA order	Seasonal order	Placebo step coefficient	p-value
Dec 2016	ARIMA(1,1,1)	(1,0,1)[12]	0.00328 (0.00224)	0.150
Feb 2017	ARIMA(1,1,1)	(1,0,1)[12]	-0.00067 (0.00223)	0.767
Mar 2017	ARIMA(1,1,1)	(1,0,1)[12]	-0.00117 (0.00225)	0.605

Placebo date	ARIMA order	Seasonal order	Placebo step coefficient	p-value
Apr 2017	ARIMA(1,1,1)	(1,0,1)[12]	-0.00147 (0.00209)	0.487
May 2017	ARIMA(1,1,1)	(1,0,1)[12]	-0.00045 (0.00284)	0.874

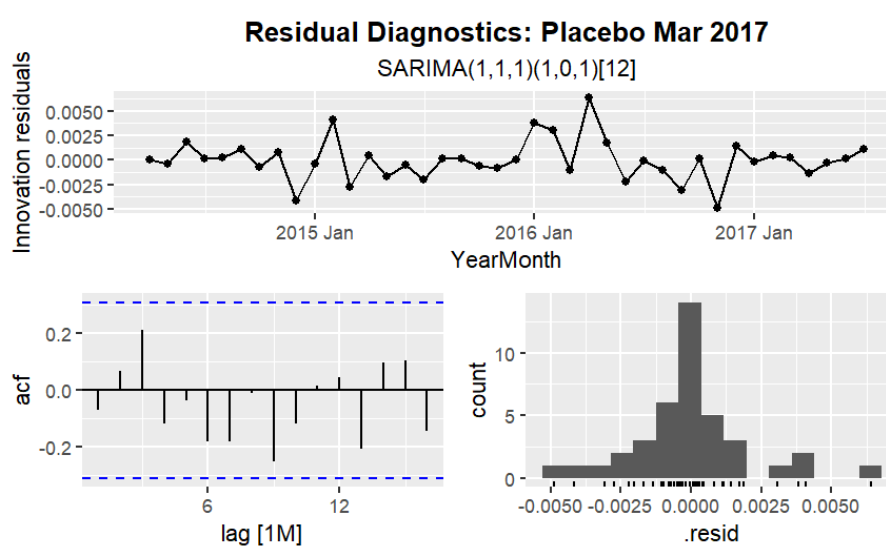
Note: Estimated coefficients are reported with standard errors in parentheses.

The in-time placebo results reported in Table 12 are reassuring. None of the successful placebo tests is statistically significant even at the 10% level. This proves that CIC/GDP was not already undergoing a statistically significant downward break before PayNow. If the baseline model were merely capturing an earlier decline in cash demand, one would expect the fake intervention dates immediately before PayNow to generate significant negative coefficients. Furthermore, the results also directly prove that the SARIMA model does not just generate intervention effects whenever a step dummy was added.

Table 13: Ljung-Box Residual Diagnostic Test for In-Time Placebo Models

Placebo date	ARIMA order	Seasonal order	Ljung-Box statistic
Dec 2016	ARIMA(1,1,1)	(1,0,1)[12]	3.1075 (p = 0.540)
Feb 2017	ARIMA(1,1,1)	(1,0,1)[12]	5.3442 (p = 0.254)
Mar 2017	ARIMA(1,1,1)	(1,0,1)[12]	6.4713 (p = 0.167)
Apr 2017	ARIMA(1,1,1)	(1,0,1)[12]	5.6791 (p = 0.224)
May 2017	ARIMA(1,1,1)	(1,0,1)[12]	5.3376 (p = 0.254)

Note: p-values are presented in parentheses.

Figure 10: Visual Residual Diagnostic Tests for March 2017 In-Time Placebo Model.⁹

From Table 13 and Figure 10, the residual diagnostic results are also satisfactory. For all successful placebo models, the Ljung-Box tests also fail to reject the null hypothesis of no residual autocorrelation. The p-value ranges from 0.167 to 0.540. This means that the placebo models are not failing because of unresolved residual dependence. Overall, the placebo analysis provides strong evidence that the results from the baseline intervention model are not driven by spurious model behaviour or the presence of structural breaks before the release of PayNow, thus strengthening the findings.

Negative-Control Outcome Falsification Test

To further assess if the results from the baseline-intervention model are truly credible, a falsification test is conducted. Theoretically, PayNow should affect variables plausibly connected to domestic cash demand but should not produce a structural break in an unrelated macro-financial outcome. Hence, if the same SARIMA baseline-intervention

⁹ Visual residual diagnostic plots for the remaining placebo models are reported in Appendix B owing to space constraints.

model found a significant PayNow effect in a variable that should not respond to PayNow, this would suggest that the result may be capturing a broader macroeconomic shock or yet another mechanical feature of the baseline intervention model specification rather than the specific effect of payment innovation.

The negative-control outcome used for this test is monthly SGD against USD exchange rate. PayNow is not expected to directly alter exchange rate during and after its launch because it is a domestic retail payment innovation rather than a change in monetary policy, capital-flow conditions, or Singapore’s external trade position.

Table 14: Coefficient Estimates for FX Negative-Control Outcome SARIMA Model

Model	ARIMA order	Seasonal order	PayNow step	p-value
fx_step_model	ARIMA(1,1,1)	(1,0,1)[12]	-0.00262 (0.01442)	0.856

Note: Estimated coefficients are reported with standard errors in parentheses.

The negative-control outcome estimates in Table 14 provide little evidence of a PayNow-related break in the SGD/USD exchange-rate series. Applying the same step_111_101 specification, the estimated PayNow step coefficient is −0.00262, with a standard error of 0.01442 and a p-value of 0.856. This coefficient is statistically insignificant, indicating that the August 2017 intervention structure does not detect a comparable break in an outcome that should not plausibly respond to PayNow. If the baseline-intervention model result were caused by the model mechanically imposing a break around August 2017, or by an economy-wide event in that month affecting many

macroeconomic variables, one expects the same intervention structure to detect an effect in the exchange rate.

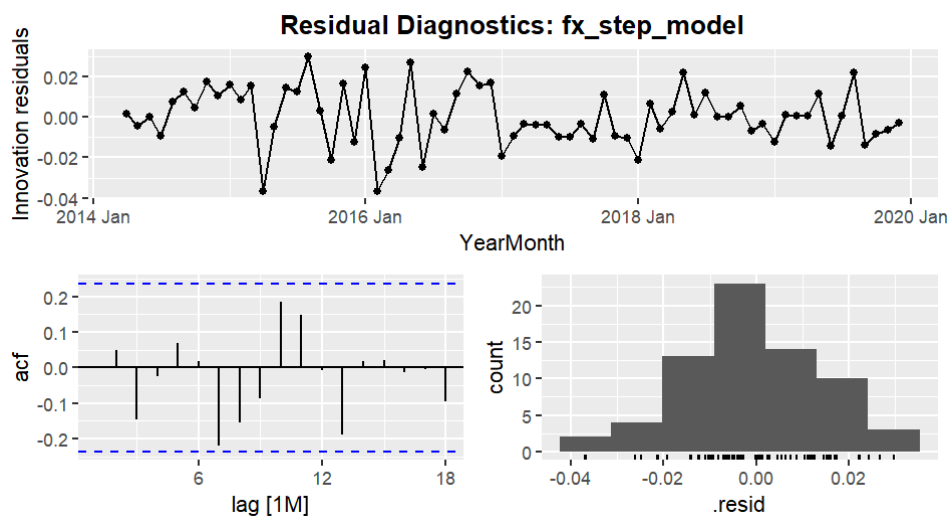
Table 15: Ljung-Box Residual Diagnostic Test for FX Negative-Control Outcome

SARIMA Model

Model	ARIMA order	Seasonal order	Ljung-Box statistic
fx_step_model	ARIMA(1,1,1)	(1,0,1)[12]	11.2865 (p = 0.080)

Note: p-values are presented in parentheses.

Figure 11: Visual Residual Diagnostic Tests for the SGD/USD Negative-Control Outcome SARIMA Model.



The residual diagnostic results provided by Table 15 and Figure 11 for the negative-control outcome model are also reassuring. Since exchange rate is not expected to exhibit strong annual seasonality (Cellini and Cuccia, 2011), the Ljung-Box test lag is

set as $\min(10, \frac{T}{5})$ for the negative-control outcome falsification test (Hyndman, 2014), rather than $\min(2s, \frac{T}{5})$. The Ljung-Box statistic is 11.2865, with p-value 0.080, meaning that the null hypothesis of no residual autocorrelation cannot be rejected at the conventional 5% significance level. Hence, the absence of a negative-control outcome effect is not due to a poorly fitted model leaving substantial serial dependence in the residuals. Therefore, the negative-control outcome results reduce concern that the significant CIC/GDP results reflect a general macro-financial disruption or a mechanical tendency for the baseline-intervention model to identify a structural break in August 2017. This significantly strengthens the results of the step_111_101 model.

8. APPENDIX B

Figure 12: Visual Residual Diagnostic Tests for December 2016 In-Time Placebo

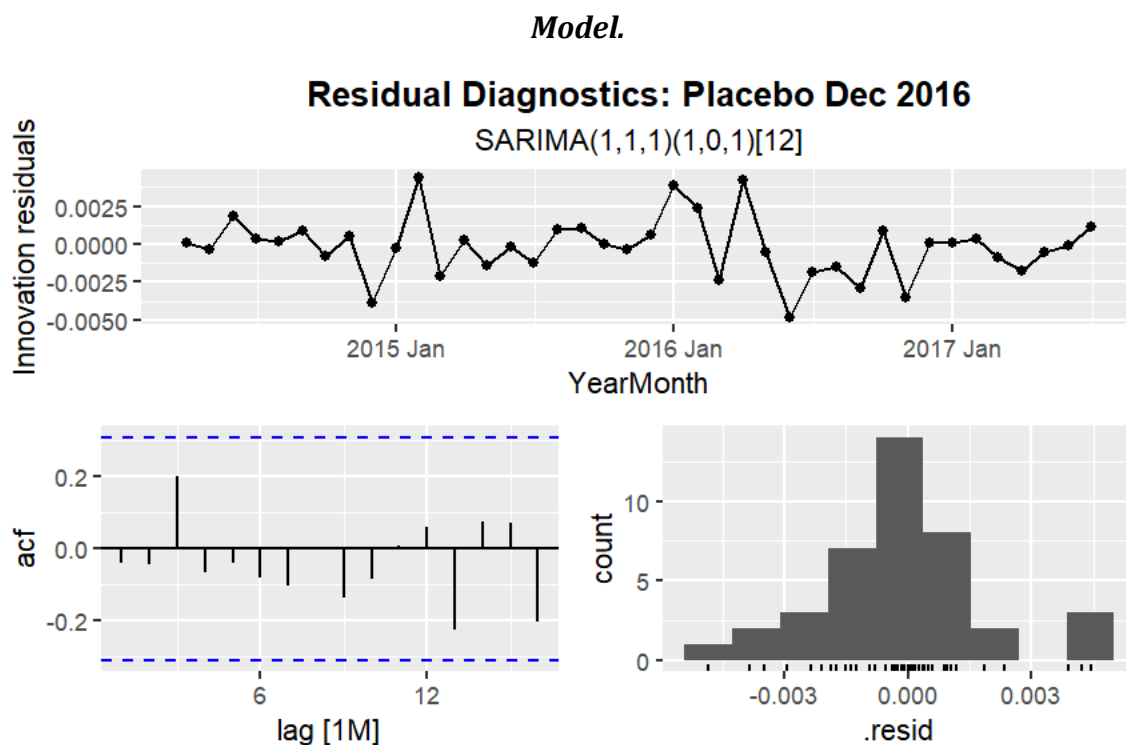


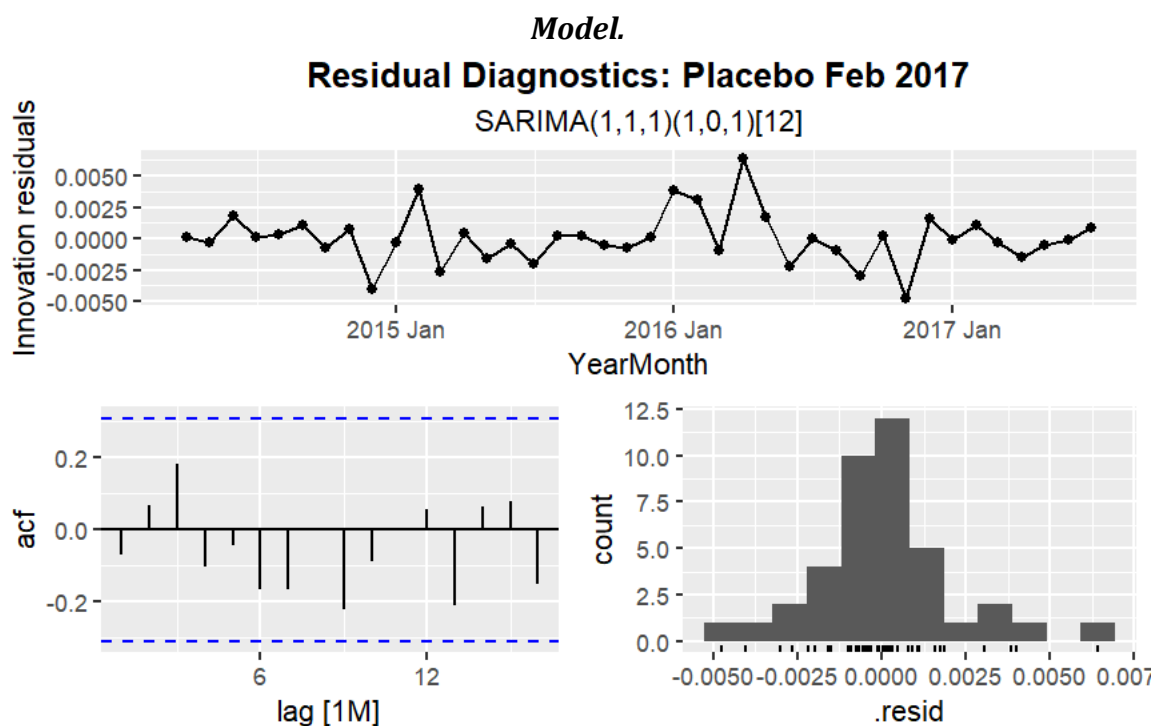
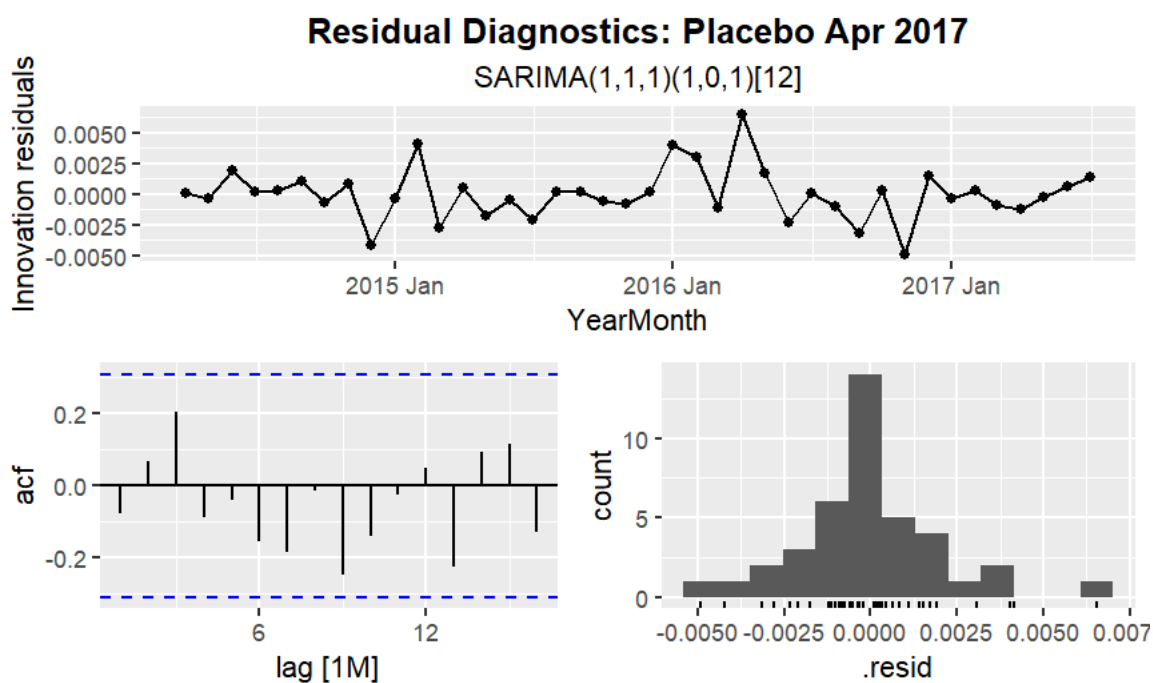
Figure 13: Visual Residual Diagnostic Tests for February 2017 In-Time Placebo**Figure 14: Visual Residual Diagnostic Tests for April 2017 In-Time Placebo Model.**

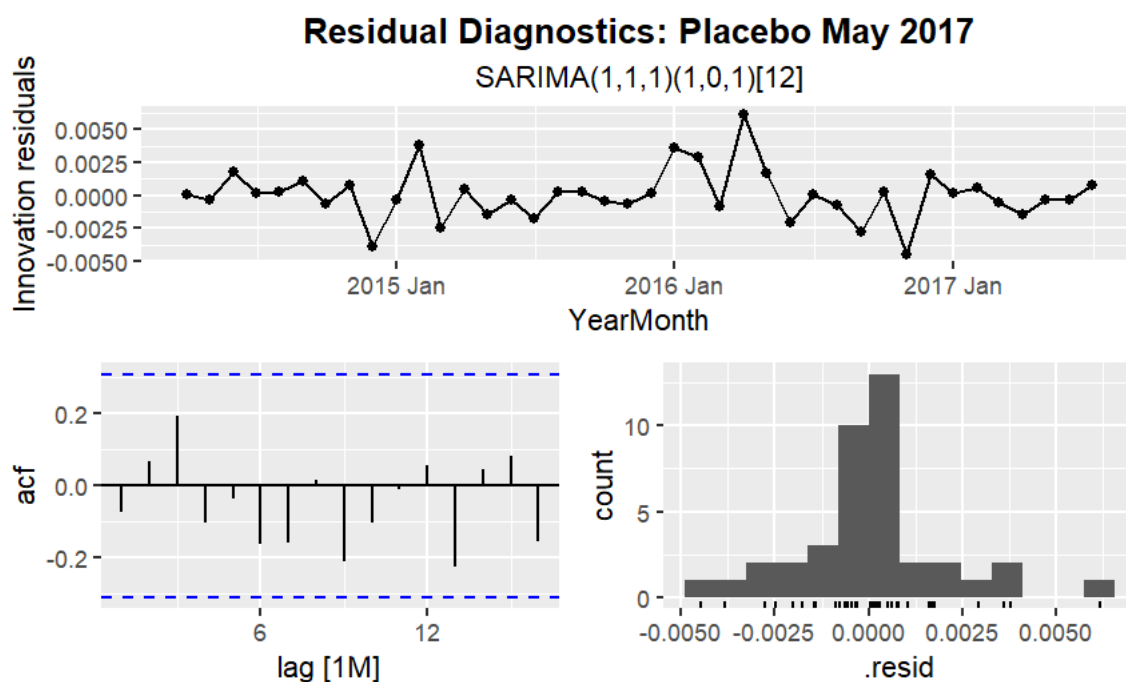
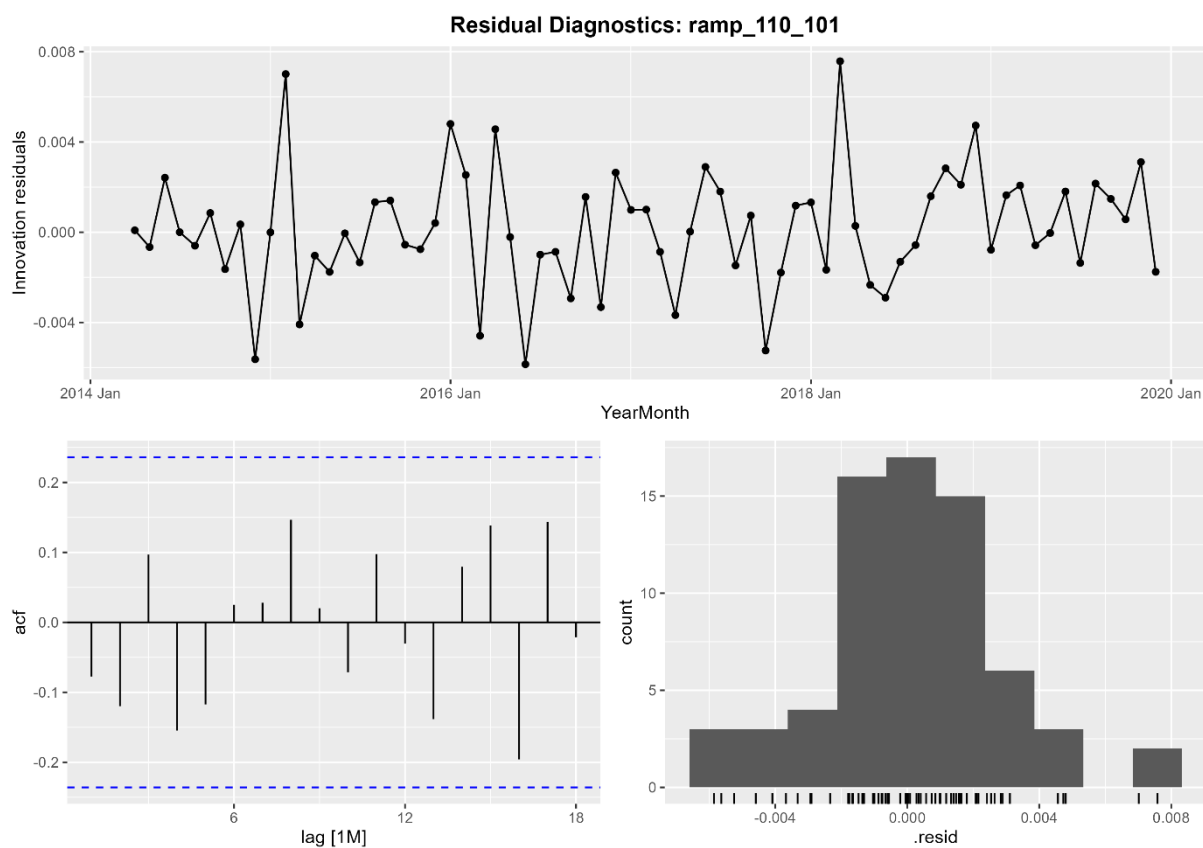
Figure 15: Visual Residual Diagnostic Tests for May 2017 In-Time Placebo Model.**Figure 16: Visual Residual Diagnostic Tests for ramp_110_101 Model.**

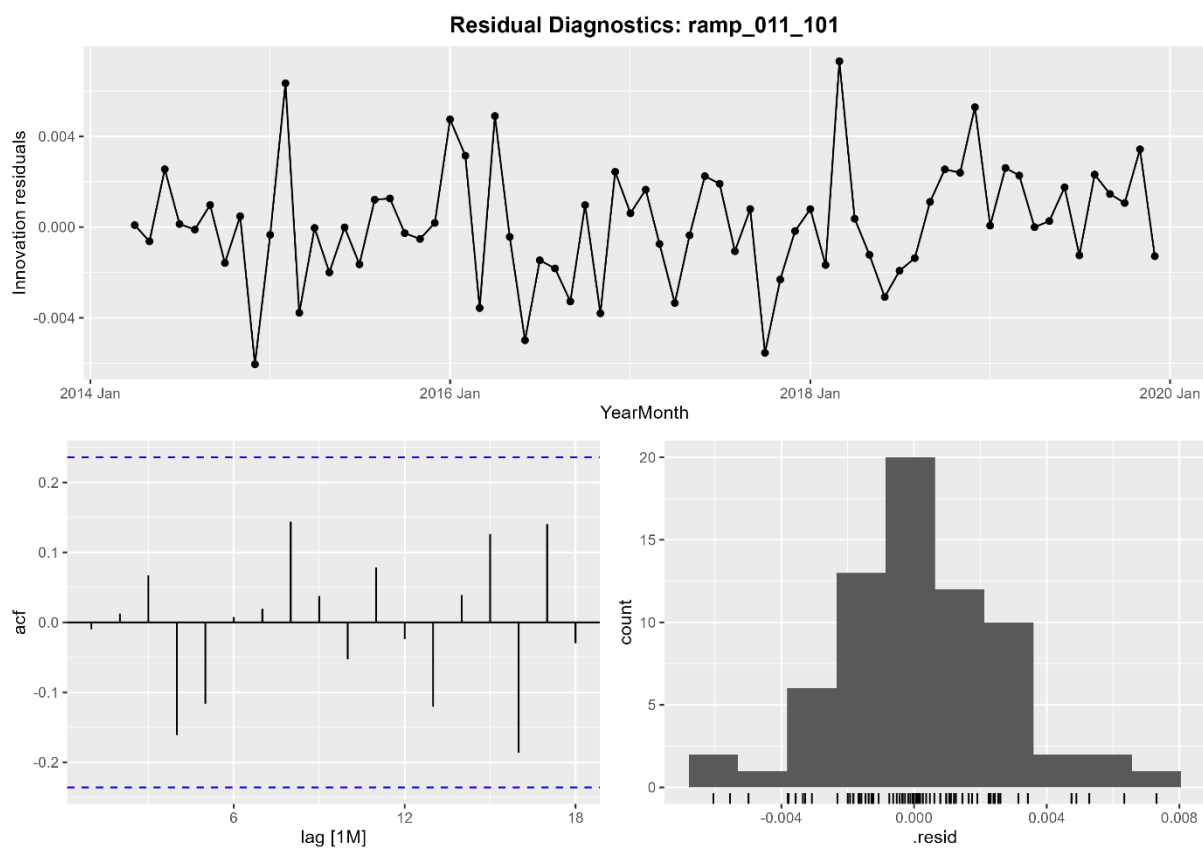
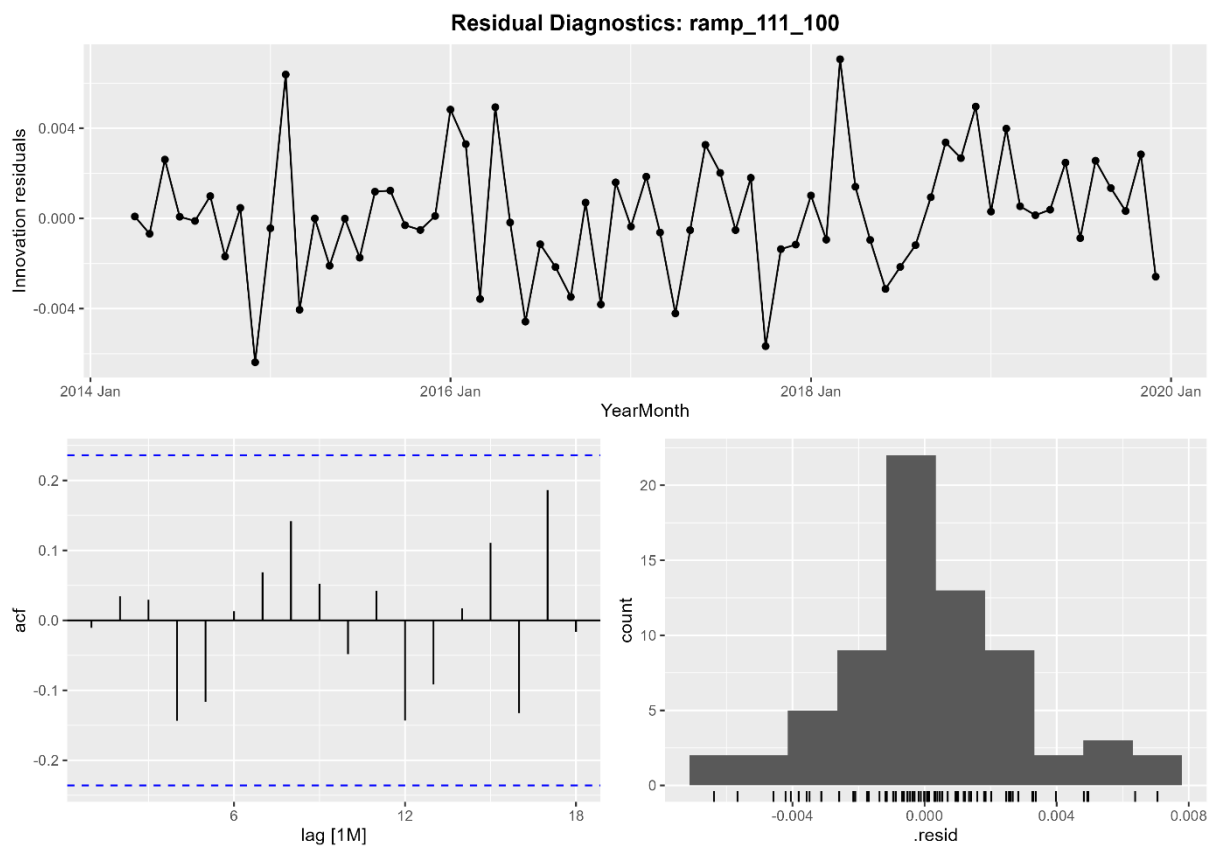
Figure 17: Visual Residual Diagnostic Tests for ramp_011_101 Model.**Figure 18: Visual Residual Diagnostic Tests for ramp_111_100 Model.**

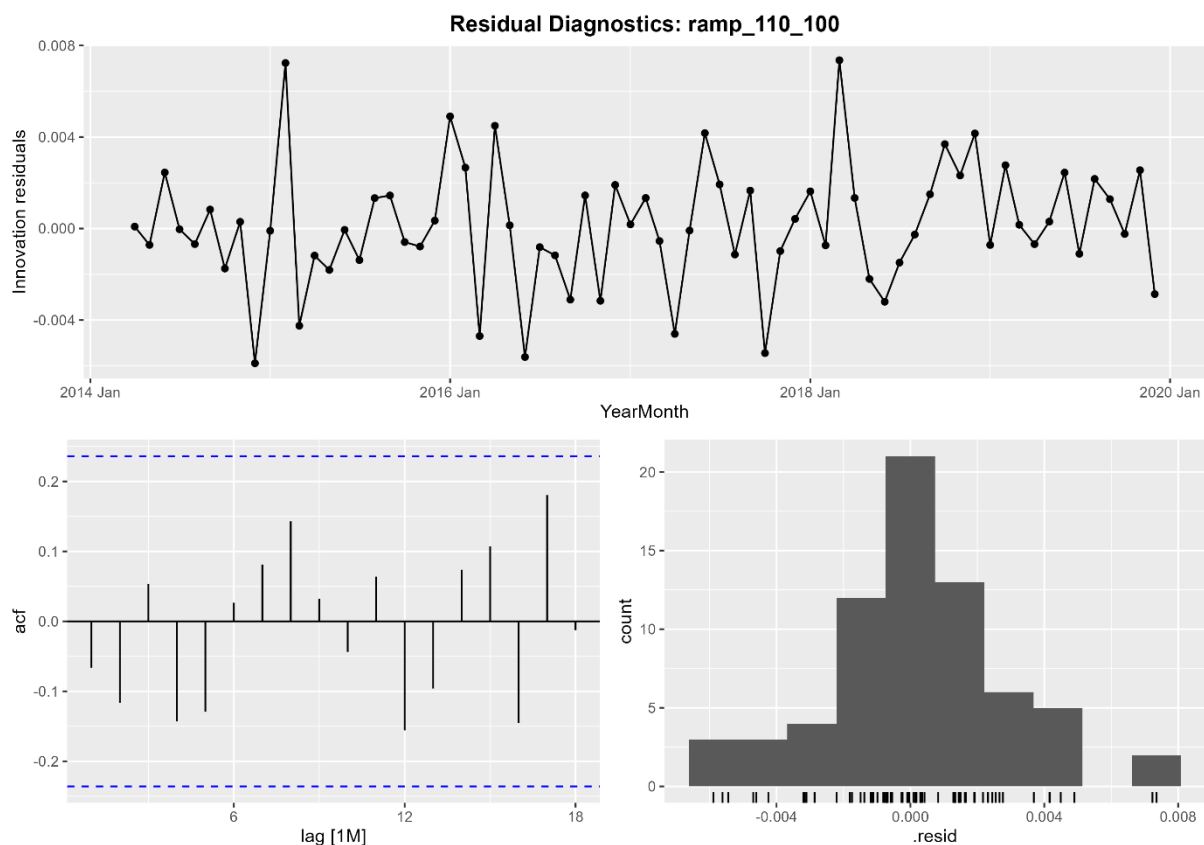
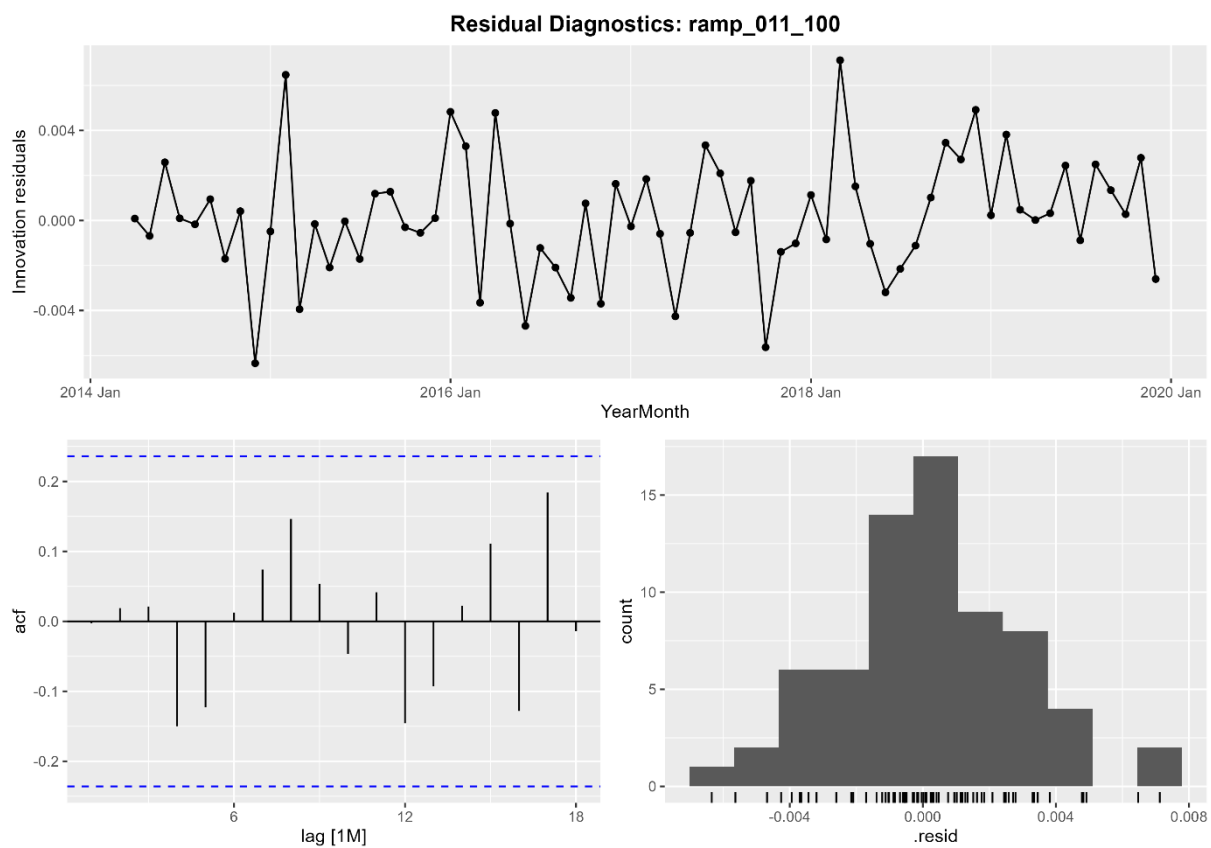
Figure 19: Visual Residual Diagnostic Tests for ramp_110_100 Model.**Figure 20: Visual Residual Diagnostic Tests for ramp_011_100 Model.**

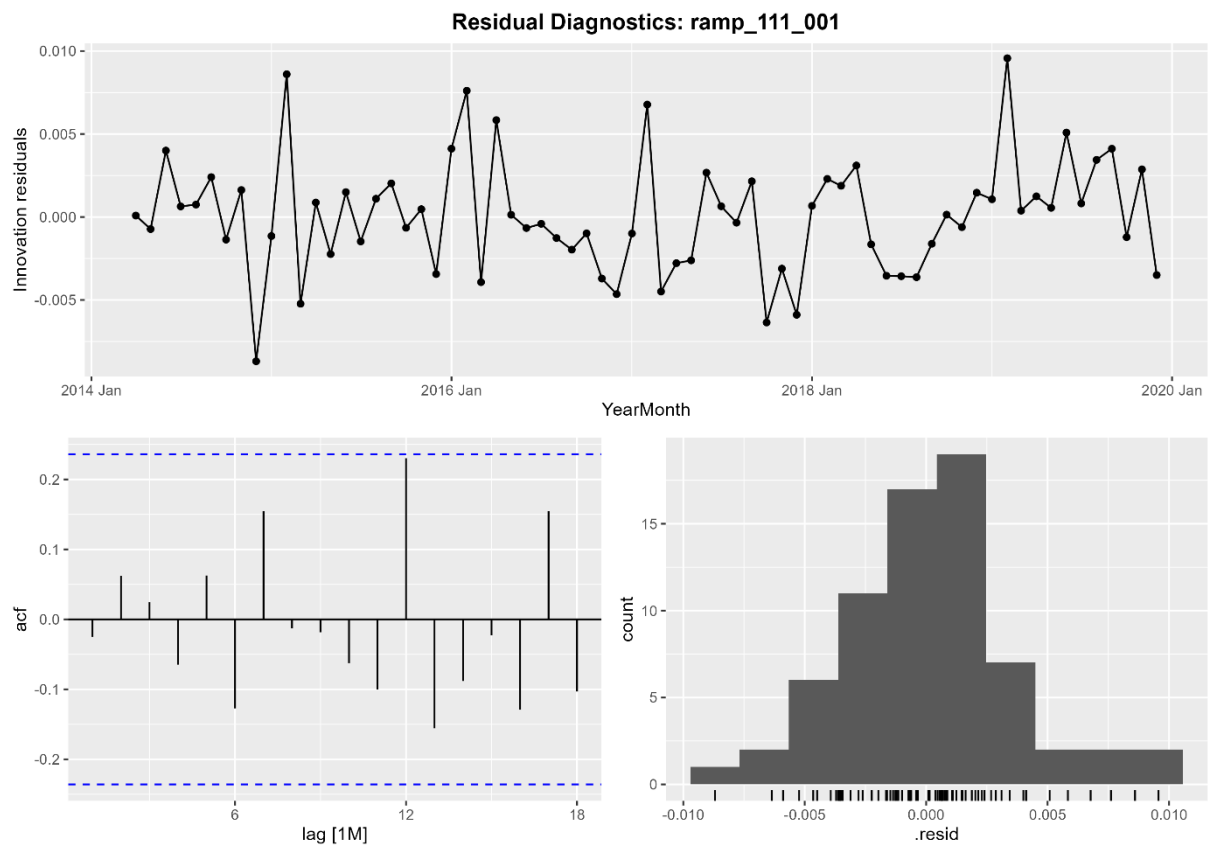
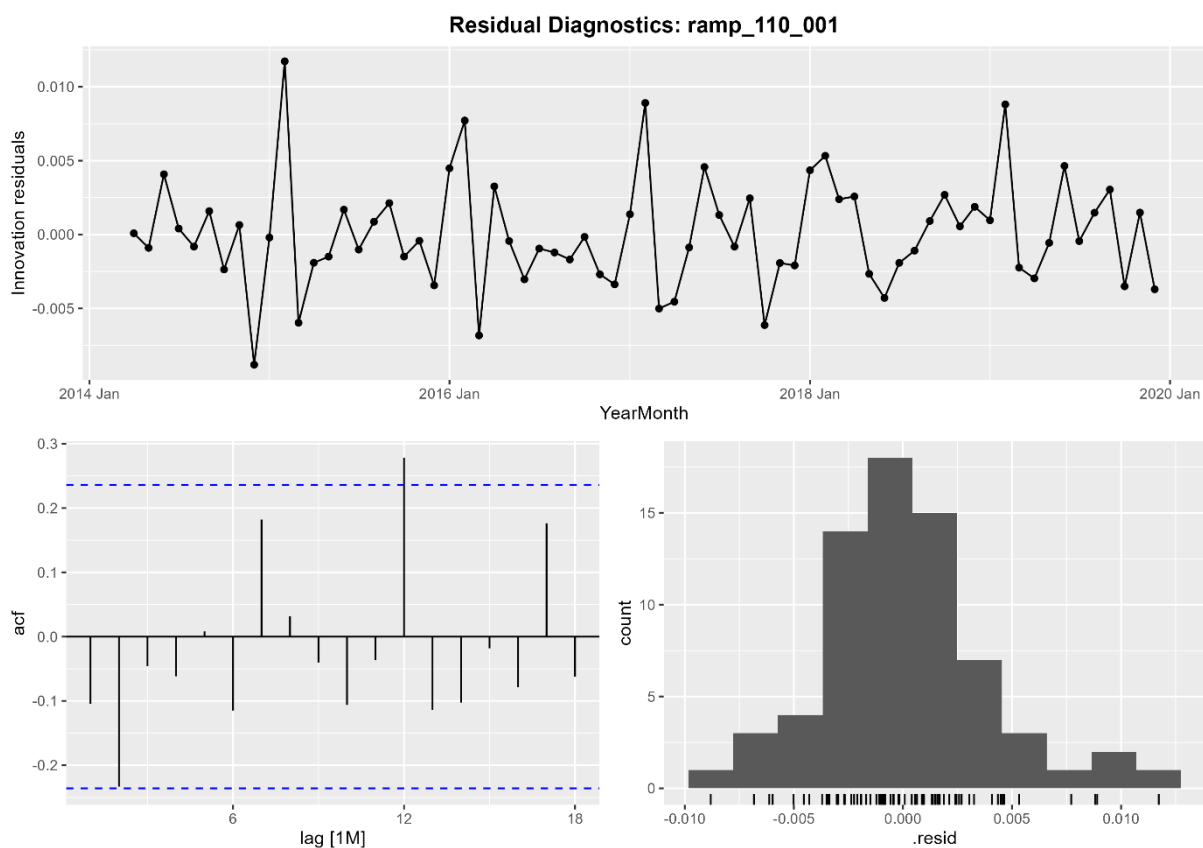
Figure 21: Visual Residual Diagnostic Tests for ramp_111_001 Model.**Figure 22: Visual Residual Diagnostic Tests for ramp_110_001 Model.**

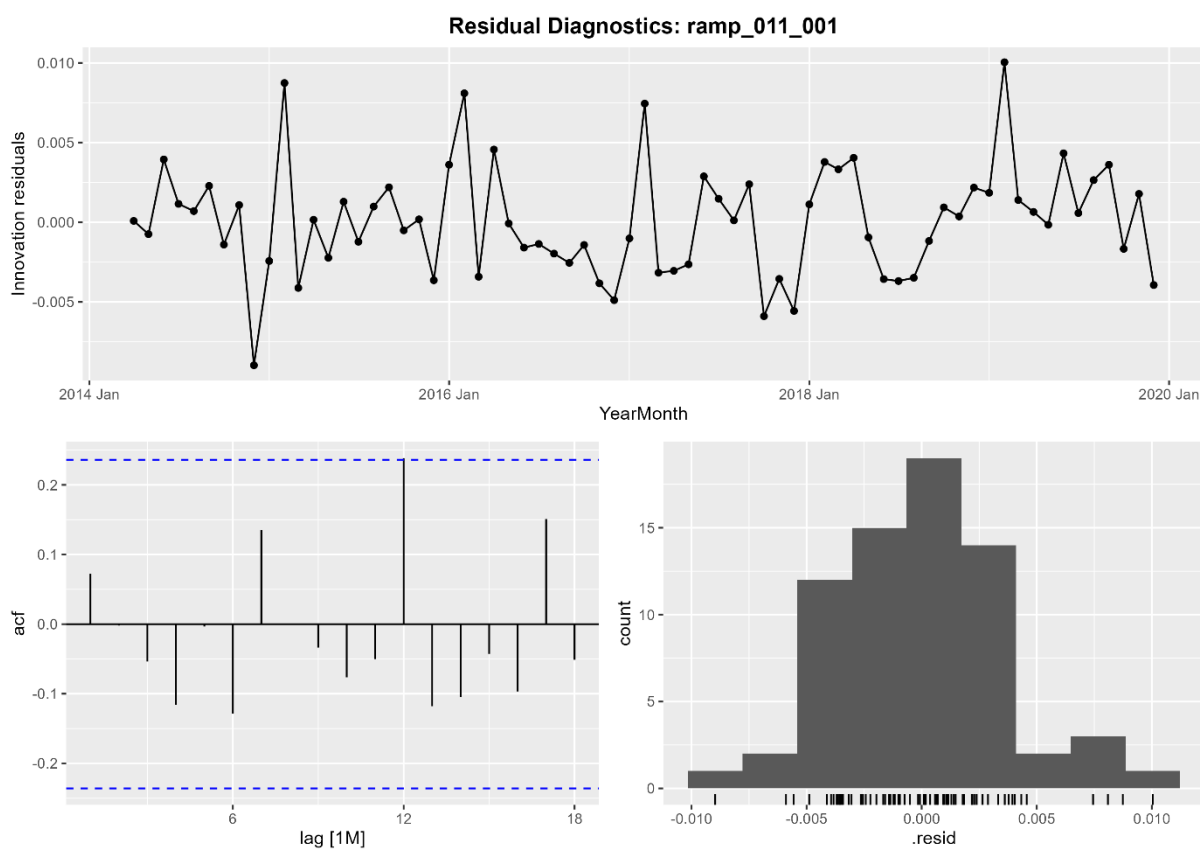
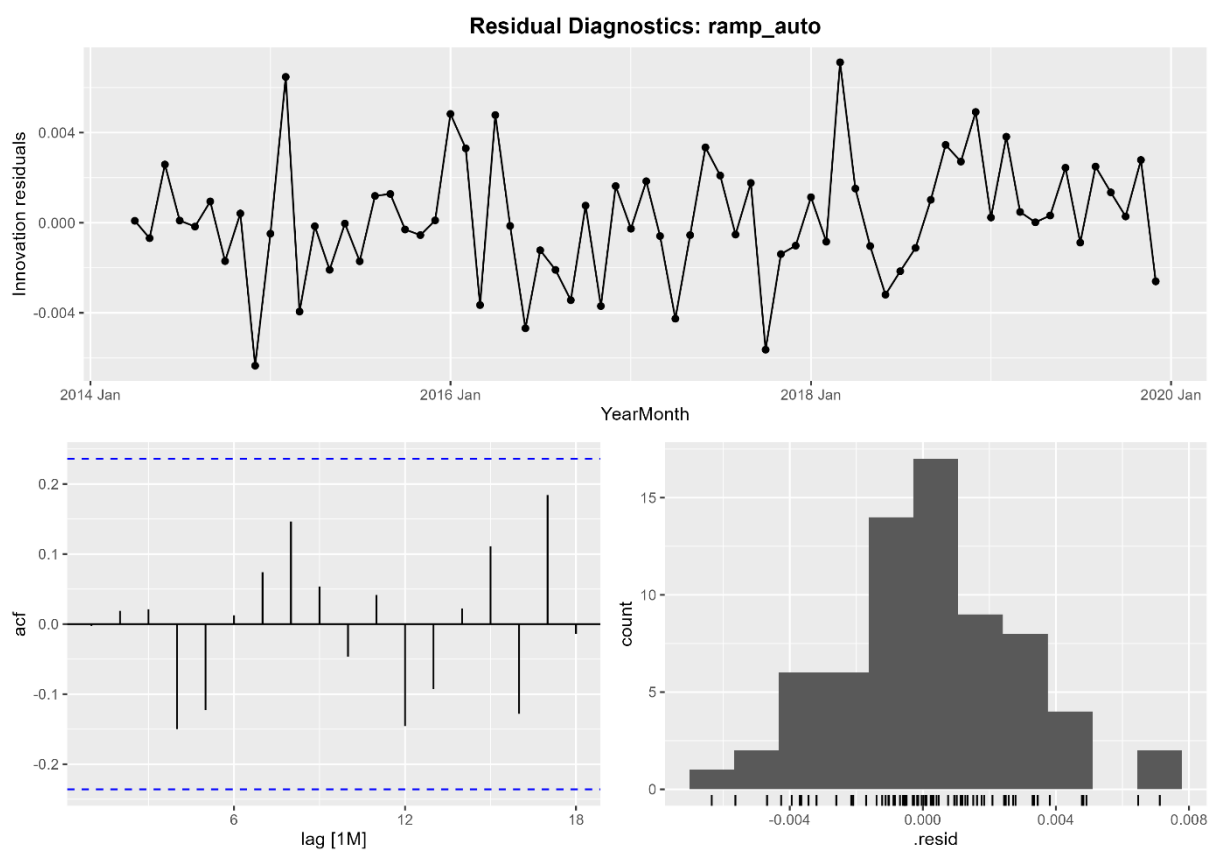
Figure 23: Visual Residual Diagnostic Tests for ramp_011_001 Model.**Figure 24: Visual Residual Diagnostic Tests for ramp_auto Model.**

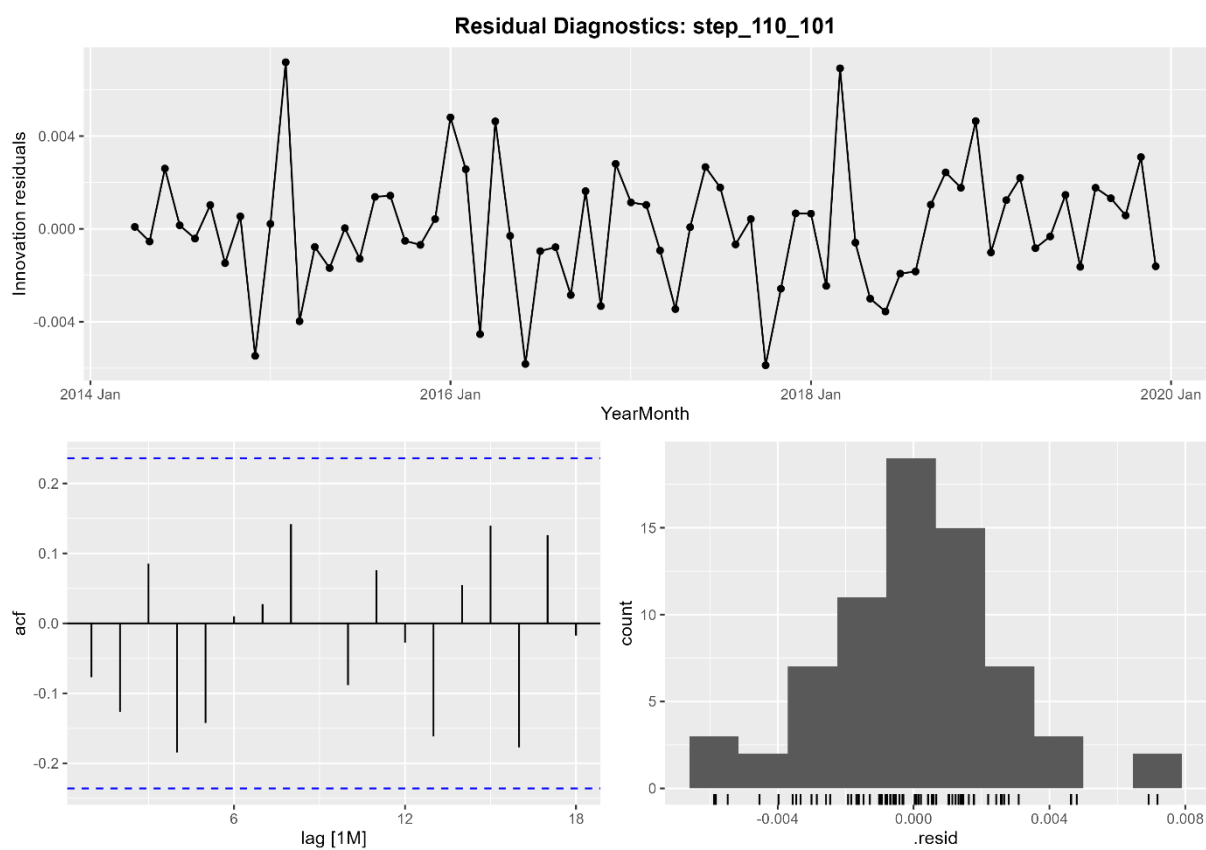
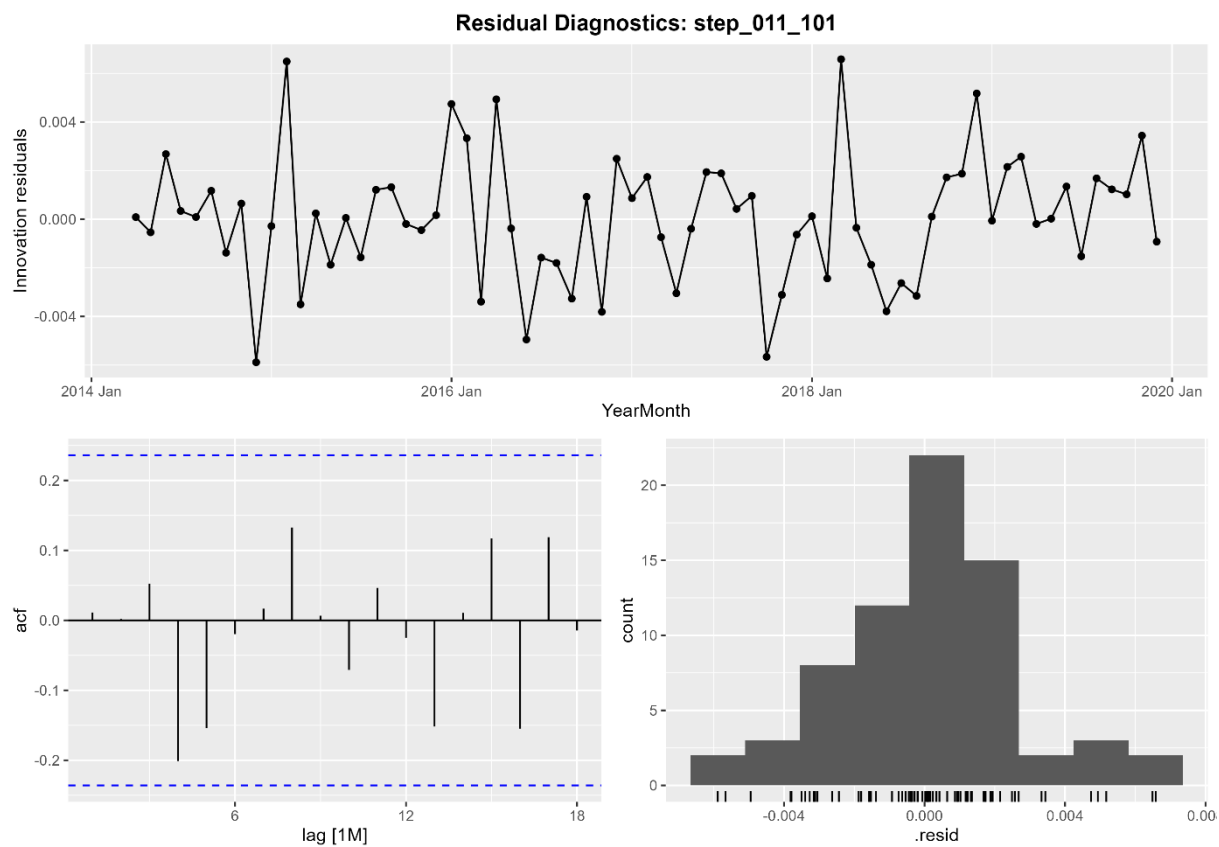
Figure 25: Visual Residual Diagnostic Tests for step_110_101 Model.**Figure 26: Visual Residual Diagnostic Tests for step_011_101 Model.**

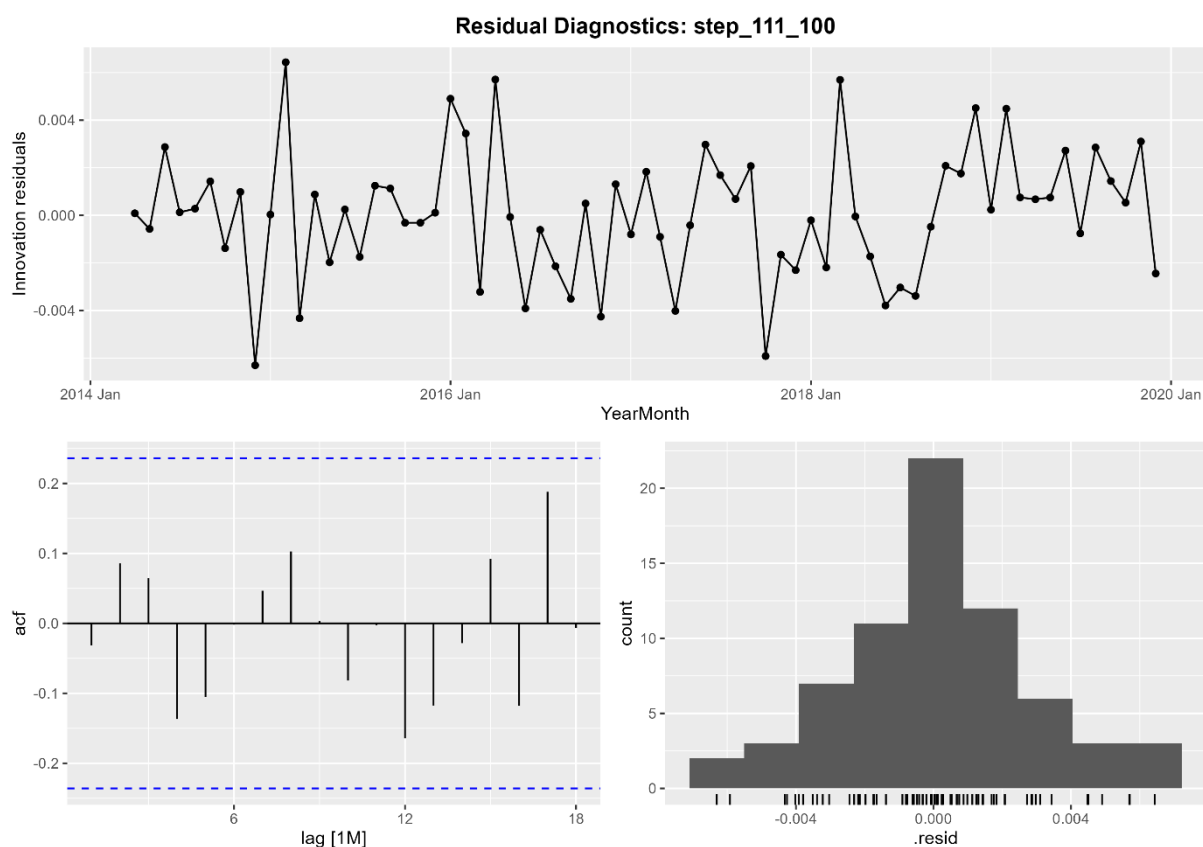
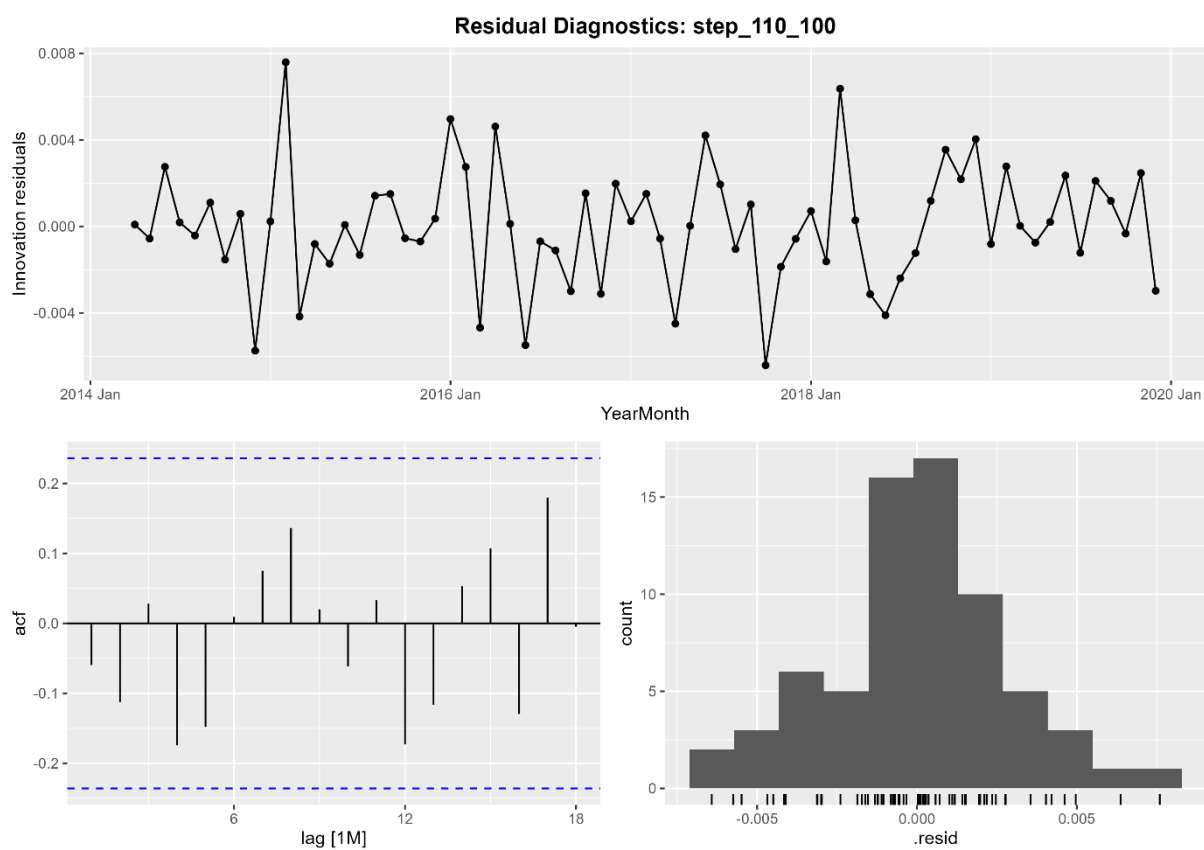
Figure 27: Visual Residual Diagnostic Tests for step_111_100 Model.**Figure 28: Visual Residual Diagnostic Tests for step_110_100 Model.**

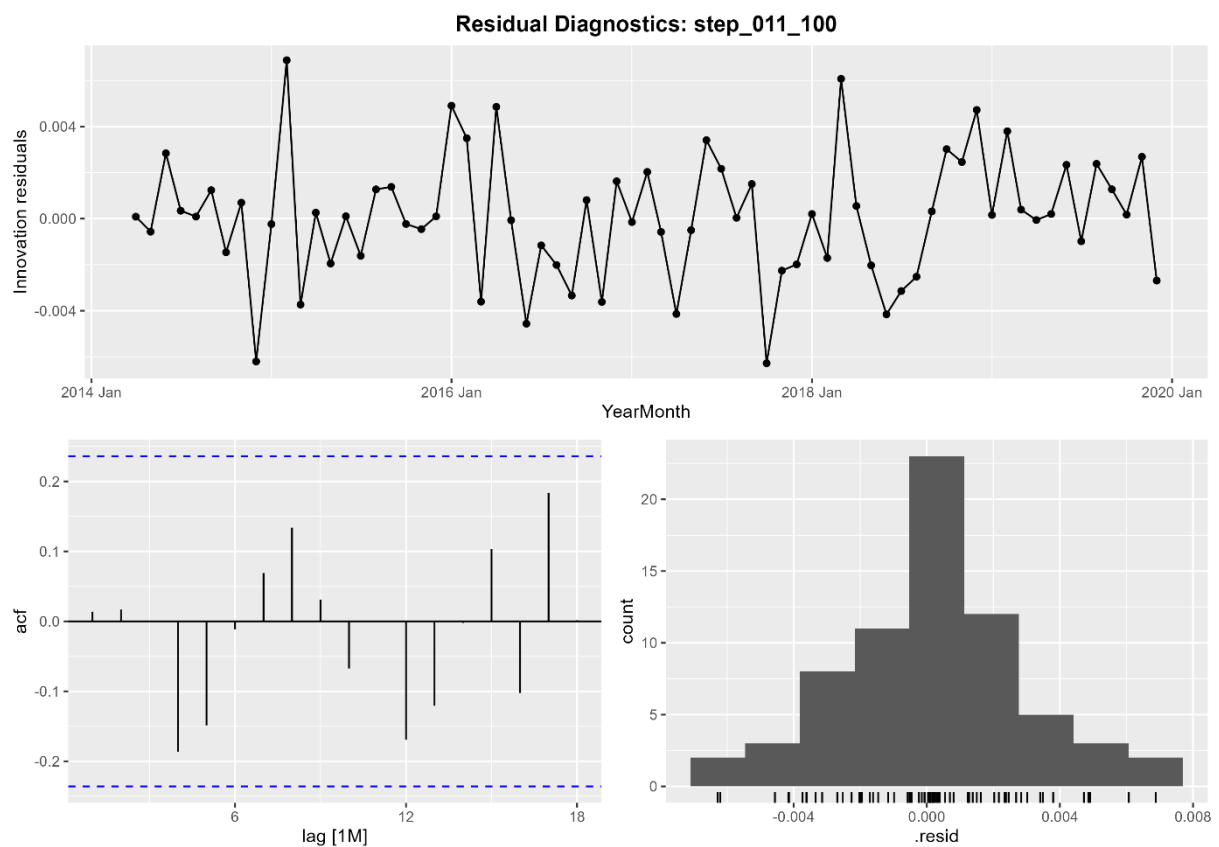
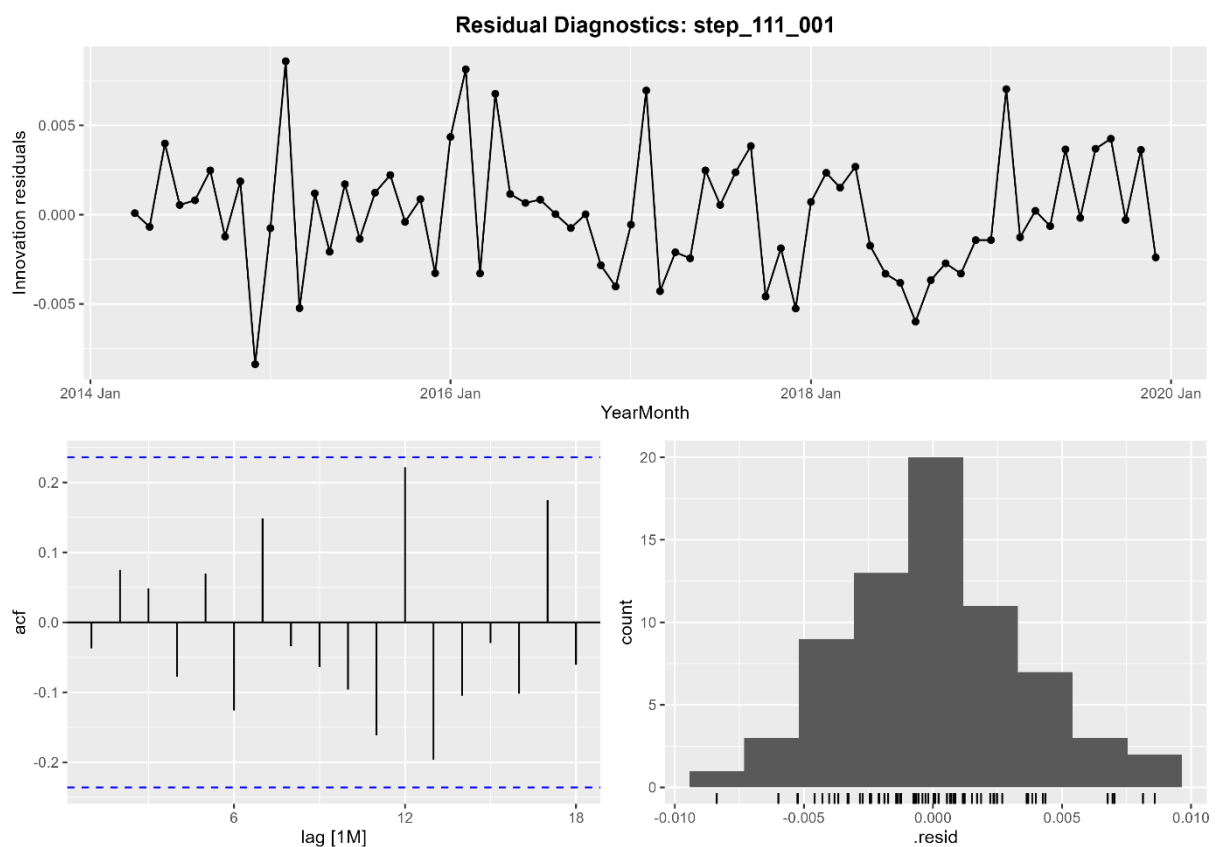
Figure 29: Visual Residual Diagnostic Tests for step_011_100 Model.**Figure 30: Visual Residual Diagnostic Tests for step_111_001 Model.**

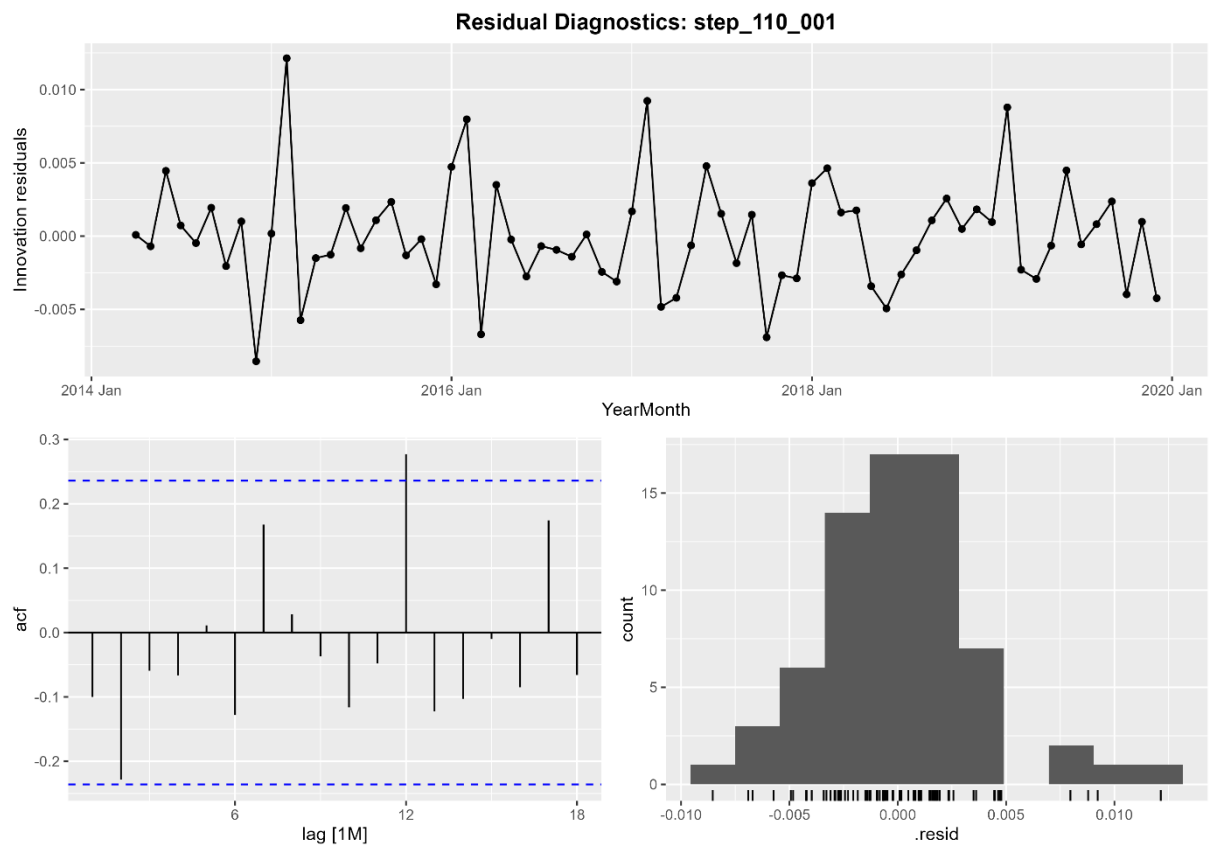
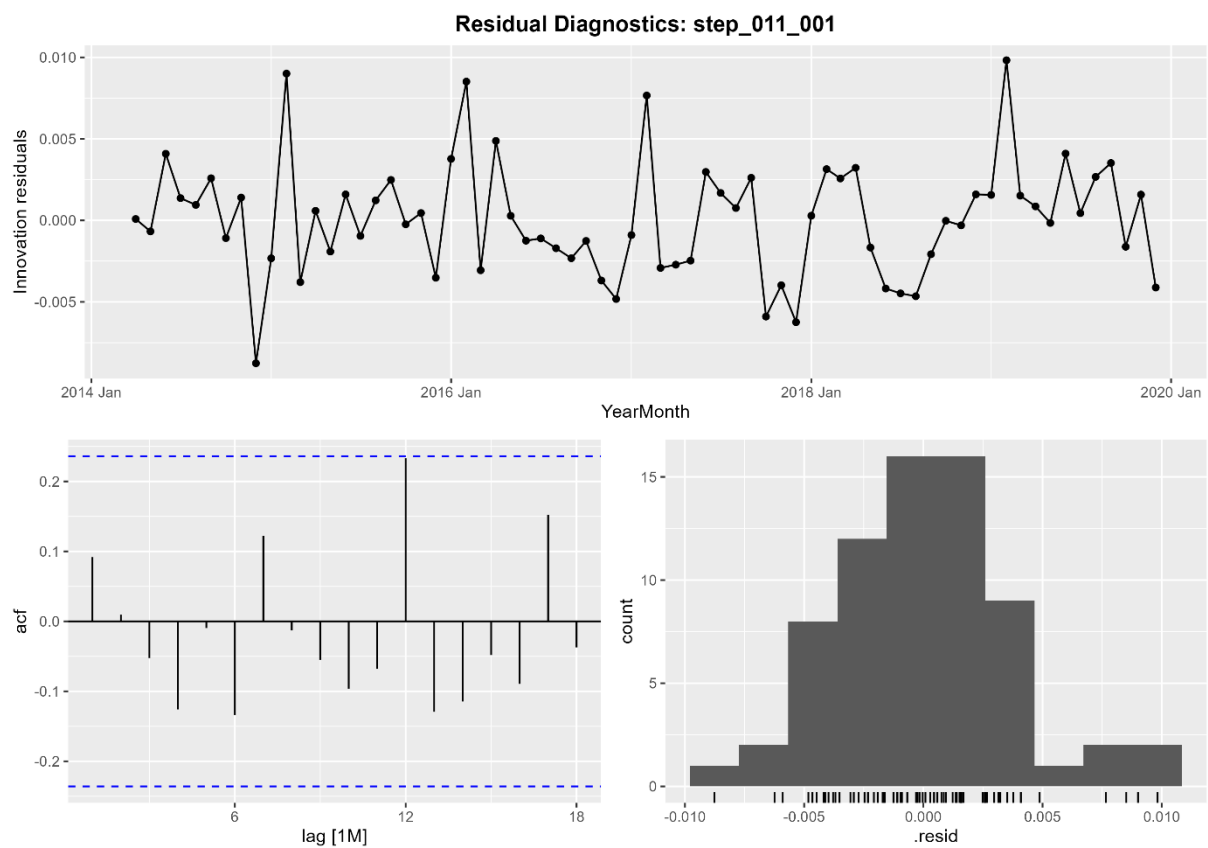
Figure 31: Visual Residual Diagnostic Tests for step_110_001 Model.**Figure 32: Visual Residual Diagnostic Tests for step_011_001 Model.**

Figure 33: Visual Residual Diagnostic Tests for step_auto Model.